

Proposal to
**Department of Culture and
Tourism (DCT)**
For
Oracle Tenant DRCC

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For further information, please contact:

Imad AL Badawi

Senior Account Manager
Core42 Technology Projects LLC

Email: imad.albadawi@core42.ai

Mobile: +971 (52) 6089501

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1 Executive Summary

Core42 is committed to driving long-term success for your organisation. With our local roots and global strength, we stand ready to deliver transformative, reliable, and aligned results that align with DCT's goals.

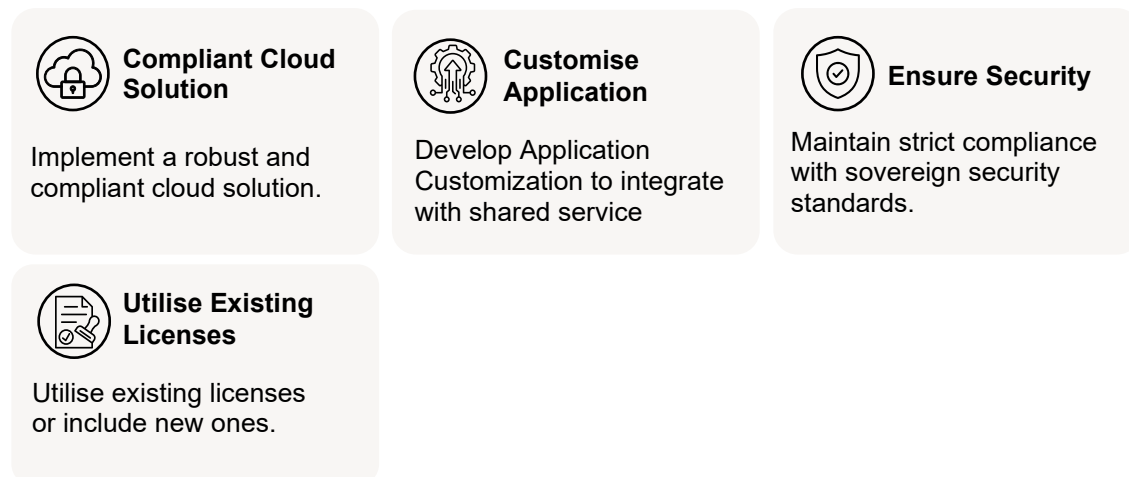
We would like to extend our sincerest appreciation to DCT for entrusting us with the opportunity to propose our Core42 Oracle Private Cloud Consumption Services, specifically designed to support developing their Oracle Applications on a secure and scalable cloud environment, Core42 Oracle Private Cloud.

1.1 Our Understanding of DCT's Objectives

◆ DCT's Objectives ◆

Our solution strengthens digital presence, boosts engagement, and modernises infrastructure for high availability, disaster recovery, and scalability. It lowers total cost of ownership, streamlines operations, and builds resilience through digital agility. Improved performance and efficiency drive loyalty, retention, and cost savings delivering stronger loyalty, retention, and cost savings through improved performance and efficiency.

◆ Understanding of DCT's Asks ◆



◆ Our Proposed Solution ◆

Core42 Oracle Private Cloud, powered by Oracle Cloud Infrastructure, offers flexible deployment options with both "Bring Your Own License" and "License Included" models. It ensures high availability and disaster recovery while maintaining compliance with UAE regulations.

Figure 1: Our Understanding of DCT's Objectives

1.2 Salient Benefits and Expected Outcomes

1.2.1 Strategic Benefits

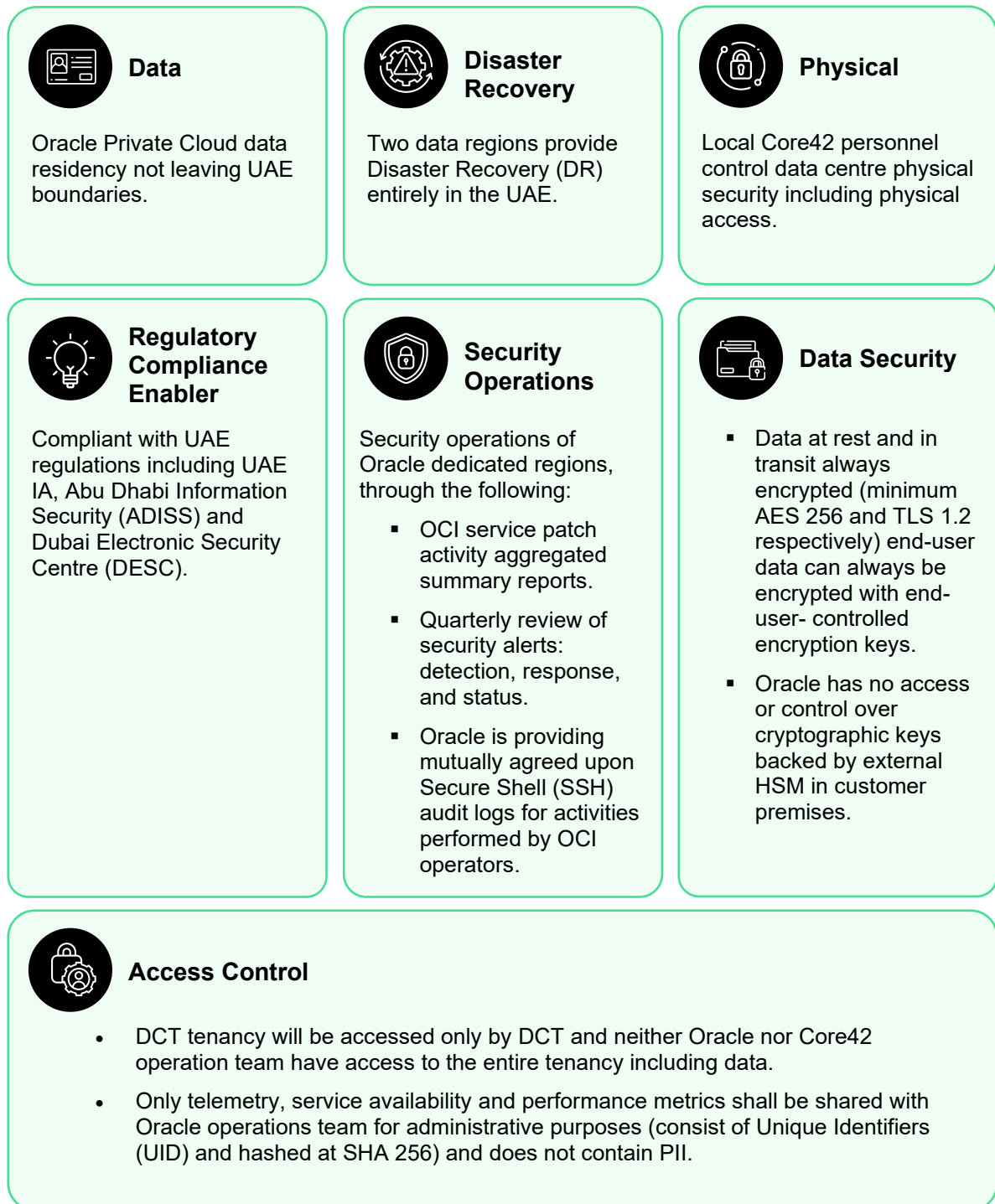


Figure 2: Strategic Benefits

1.2.2 Accelerated Digital Transformation via Partnerships

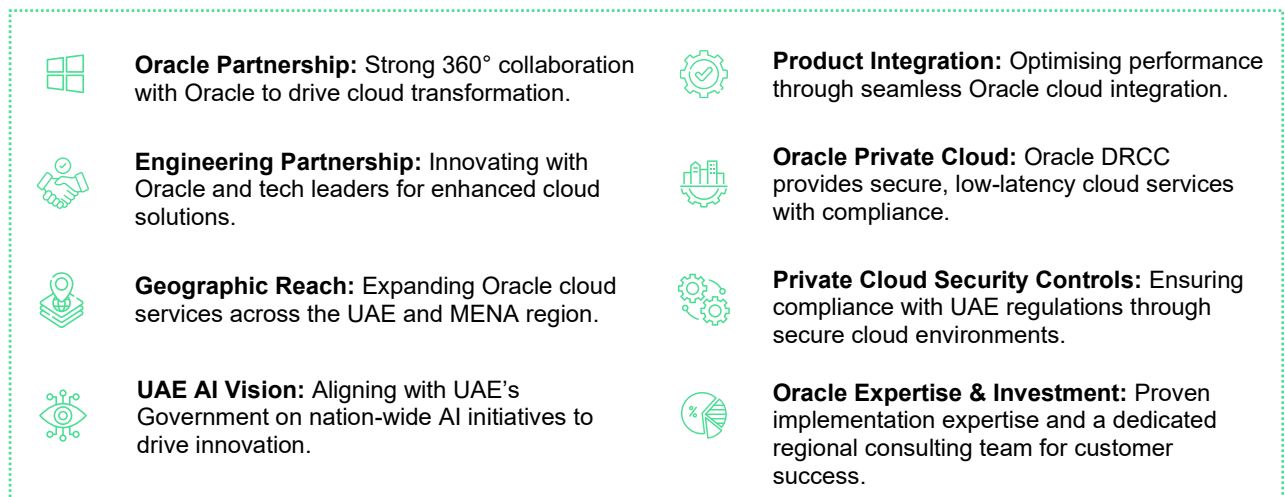


Figure 3: Accelerated Digital Transformation via Partnerships

Core42 has unique deep Oracle experience, robust value proposition, proven approach, and extensive partnership network.

1.3 Proven Execution Across Abu Dhabi Government Entities

Core42's Oracle Private Cloud platform is not a theoretical construct, it is already operational, validated, and supporting critical Abu Dhabi Government workloads today.

1.3.1 Active ADGE Deployments on Core42 Oracle Private Cloud

Two major Abu Dhabi Government Entities (DMT and ADAFSA) are already running production/DR workloads on Core42's Oracle DRCC IaaS/PaaS services. These live environments demonstrate:

- Operational stability and performance at sovereign scale.
- Compliance alignment with Abu Dhabi Government regulatory frameworks.
- Proven governance, security, and service management controls.
- Production-grade workload migration and stabilisation capabilities.

This existing footprint significantly reduces onboarding risk and accelerates adoption and integration for DCT.

1.3.2 Sovereign Multi-Cloud Disaster Recovery Architecture

Core42 has successfully implemented and operationalised **multi-cloud connectivity between Azure Sovereign Public Cloud and Core42 Oracle Private Cloud** to support DMT's Disaster Recovery strategy.

This investment delivers:

- Secure, low-latency cross-cloud connectivity.

- Sovereign-aligned DR architecture.
- Operational DR runbooks and tested failover mechanisms.
- A validated hybrid-cloud operating model.

The DR capability is already live, proving Core42's ability to design and operate sovereign multi-cloud architectures beyond single-platform deployments.

1.3.3 Integration with AD Government Strategic Platforms (ADERP 2.0)

Core42 Oracle Private Cloud currently hosts AD Government ADERP 2.0, enabling:

- Direct ecosystem integration for participating entities.
- Reduced dependency complexity during migration.
- Simplified interoperability with government-shared services.
- Alignment with Abu Dhabi's centralised ERP modernisation strategy.

For DCT, this means seamless alignment with existing government platforms without additional infrastructure re-engineering.

2 We Understand the Market Landscape



Market Drivers

Increasing Regulations:

Data residency and sovereignty necessitate public sector and regulated customers to store data within their jurisdiction, ensuring complete alignment with national and sectoral policies.

Need for Cost Optimisation:

Organisations globally are looking to reduce costs and improve efficiency while ensuring scalability and flexibility.

Desire for Modernised Infrastructure:

Most customers host heterogeneous workloads facing infrastructure end-of-life. These require high software license costs, high availability and disaster recovery solutions; thereby creating a strong desire for modernised infrastructure to support business growth.

Major Investments in Cloud Infrastructure:

Core42, in collaboration with Oracle, has invested in cloud and AI infrastructure to launch two dedicated cloud regions. These regions replicate public cloud capabilities while ensuring locality and sovereignty for government and public sector entities, addressing the increasing demand for AI and cloud services.

Rapid Growth in AI:

- Adoption across industries, including healthcare, finance, retail, and manufacturing.
- Application cases include computer vision, GenAI, natural language processing, and analytics.
- Advancement in algorithms and models is fuelling the growth towards Artificial General Intelligence (AGI).



Strategic Implications

- Enhancing market position, improving DCT| satisfaction, and increasing operational efficiency are key to securing a competitive edge and fostering long-term success.
- Strengthen compliance and security measures and achieve sustainable cost savings to ensure resilience and support strategic growth.
- Accelerate decision-making with pre-built analytics and AI-driven insights.
- Designed to adapt to organisations of all sizes, offering flexibility to meet evolving business needs.
- Ensure peace of mind with industry-leading data protection and compliance capabilities.

Strategic Partnerships for Sectoral Transformation:

Oracle has partnered with the Dubai AI campus and the UAE Ministry of AI to drive innovation in healthcare, smart cities, and public services, demonstrating its commitment to regional transformation.

Sovereign Cloud Infrastructure for Scalable AI Solutions:

Oracle's sovereign cloud initiatives provide scalable AI solutions, enabling secure and efficient regional capabilities for public and private sector organisations.

3 Our Understanding of Your Requirements

Core42 understands DCT aims to subscribe the Cloud resources. Core42 to provision new DCT tenant and handover for DCT IT to Manage the Tenancy and underlying Cloud administration and Application customization.

The Core42 Oracle Private Cloud platform provides on-demand user services, compute, storage, databases, analytics, networking, mobile, developer tools, management tools, security, and enterprise applications, with a subscription-based cloud-hosted service.

4 Proposed Solution

4.1 List of Offerings

Core42 offers the following broadly scoped services across all Core42 Oracle Private Cloud:

- Oracle IaaS/PaaS Hosting Services.

Part	Description	IaaS/PaaS	Part Qty	Instance Qty	Usage Qty
	Prod Autonomous				
B95706	Oracle Autonomous AI Database Storage for Transaction Processing (Gigabyte Storage Capacity Per Month)	PaaS	150	1	1
B95754	Oracle Autonomous AI Database Storage (Gigabyte Storage Capacity Per Month)	PaaS	450	1	1
B95702	Oracle Autonomous AI Transaction Processing - ECPU (ECPU Per Hour)	PaaS	7.25	1	744
	Testing+Dev				
B95706	Oracle Autonomous AI Database Storage for Transaction Processing (Gigabyte Storage Capacity Per Month)	PaaS	300	1	1
B95754	Oracle Autonomous AI Database Storage (Gigabyte Storage Capacity Per Month)	PaaS	600	1	1
B95702	Oracle Autonomous AI Transaction Processing - ECPU (ECPU Per Hour)	PaaS	3.625	1	744
	Application Integration Main				
B89639	Oracle Integration Cloud Service - Enterprise (5K Messages Per Hour)	PaaS	1	1	744
	Application Integration Dev&Test				
B89639	Oracle Integration Cloud Service - Enterprise (5K Messages Per Hour)	PaaS	1	1	744

Table 1: List of Offerings

Part	Description	IaaS/PaaS	Part Qty	Instance Qty	Usage Qty
	Cloud Guard				
N/A	Oracle Cloud Guard (N/A)	IaaS	1	1	1
	OCI Identity and Access Management				
B93493	Identity and Access Management - External User (User Per Month)	IaaS	100	1	1
B93494	Identity and Access Management - Oracle Apps Premium (User Per Month)	IaaS	1	1	1
B93495	Identity and Access Management - Premium (User Per Month)	IaaS	100	1	1
B93496	Identity and Access Management - SMS (1 SMS Message Sent)	IaaS	1000	1	1
B93497	Identity and Access Management - Token (Token)	IaaS	10000	1	1
	Prod-Virtual Cloud Networks				
	Network Firewall				
B95403	Network Firewall Instance (Instance Per Hour)	Others	1	1	744
B95404	Network Firewall Data Processing (Gigabyte (GB) of Data Processed)	Others	10240	1	1
	Monitoring				
B90925	Monitoring - Ingestion (Million Datapoints)	IaaS	500	1	1
	Monitoring - Ingestion (Million Datapoints)	IaaS	49500	1	1
B90926	Monitoring - Retrieval (Million Datapoints)	IaaS	1000	1	1
	Monitoring - Retrieval (Million Datapoints)	IaaS	9000	1	1
	Logging				
B92593	OCI - Logging - Storage (Gigabyte Log Storage Per Month)	IaaS	10	1	1
	OCI - Logging - Storage (Gigabyte Log Storage Per Month)	IaaS	4990	1	1

Part	Description	IaaS/PaaS	Part Qty	Instance Qty	Usage Qty
	Load Balancer				
B93030	Load Balancer Base (Load Balancer)	IaaS	1	1	744
	Load Balancer Base (Load Balancer)	IaaS	1	1	2232
B93031	Load Balancer Bandwidth (Mbps Per Hour)	IaaS	1	1	7440
	Load Balancer Bandwidth (Mbps Per Hour)	IaaS	1	1	736560
	Web Application Firewall (WAF)				
B94579	Web Application Firewall - Instance (Instance Per Month)	IaaS	1	1	1
B94277	Web Application Firewall - Requests (1,000,000 Incoming Requests Per Month)	IaaS	10	1	1
	Web Application Firewall - Requests (1,000,000 Incoming Requests Per Month)	IaaS	990	1	1

Table 2: List of Offerings

5 Scope of Work

5.1 Scope Overview

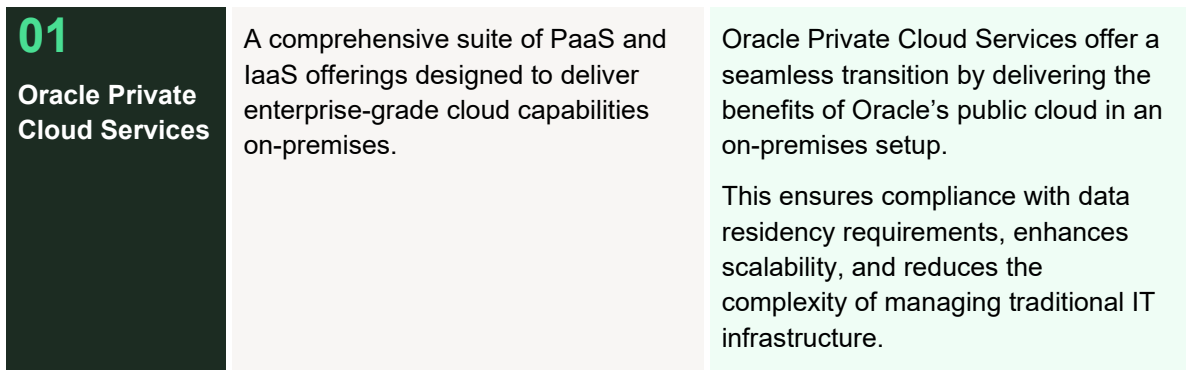


Figure 4: Scope Overview

5.2 Oracle Private Cloud

As part of this engagement, a dedicated tenant will be created for DCT on Oracle Private Cloud to establish a secure, scalable, and high-performance environment tailored to its business needs. This includes providing Oracle IaaS and PaaS Services.

To address the DCT's requirements, Core42 is proposing a solution to host DCT's Oracle Workloads on Core42's Oracle Private Cloud, which is built based on Oracle Cloud Infrastructure using the Dedicated Cloud @Customer (Oracle Private Cloud).

As the Cyber Security Council approved, this solution leverages Hyperscale cloud services for infrastructure and a platform that can be provided in-country to meet private and secure operations as per the UAE government regulations.

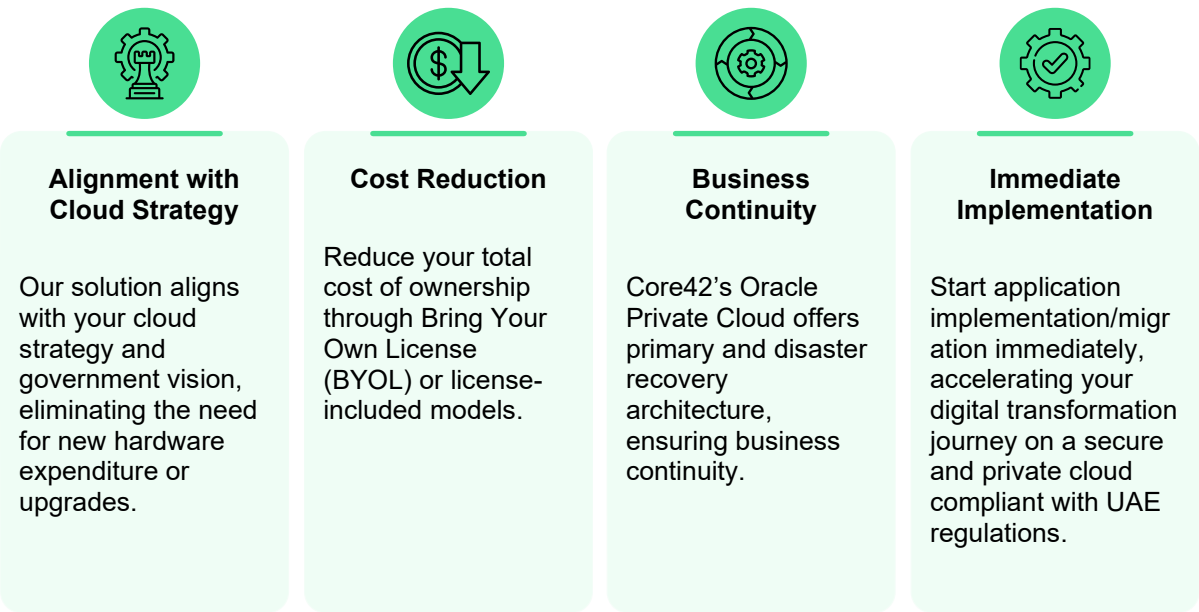


Figure 5: Scope of Work

This proposal includes the Core42's Oracle Private Cloud hosting bill of quantities targeted for consumption for 1 year.

5.3 Oracle Private Cloud (Cloud Transformation)

The scope of work outlines the deliverables, tasks, and responsibilities that will be undertaken as part of this project. It serves as a guiding framework for the project, ensuring that all activities are aligned with the agreed-upon objectives and outcomes.

5.3.1 Oracle Private Cloud Services

In alignment with our understanding of the market drivers and key industry insights, we propose a Core42's Oracle Private Cloud tailored to meet DCT's unique needs. The proposed solution leverages a secure, high-performance, and scalable environment designed to provide the flexibility and reliability required for seamless integration across DCT's infrastructure. It incorporates powerful compute options, flexible storage, and a robust networking framework, ensuring operational efficiency and system reliability.

Collaboration with Oracle

In collaboration with Oracle, Core42's Oracle Private Cloud combines the full power of Oracle's public cloud services within a private and secure environment. This solution strikes the perfect balance of scalability, flexibility, and control, making it an ideal choice for modernising DCT's infrastructure.

Powered by Oracle Cloud Infrastructure

Core42's Oracle Private Cloud is powered by Oracle Cloud Infrastructure in a dedicated region and offers the full power of the cloud services offered in a dedicated Cloud.

Managed Cloud Region

Core42's fully managed cloud region offers a flexible and scalable dedicated infrastructure optimised for diverse environments, allowing Customers to deploy infrastructure dedicated to their enterprise, in the location of their choice, with the same advantages of OCI's public cloud. With the Oracle Private Cloud, Customers can maintain control over their data and applications while benefiting from the cloud's scalability, security, and innovation.

Benefits:

1.

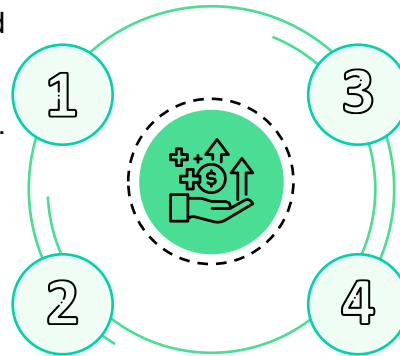
Enjoy a comprehensive, integrated cloud environment.
Address data residency and security requirements.
Enable growth and technological advancement.

2.

Delivers public cloud benefits with strict security, regulatory, and data residency compliance across Abu Dhabi regions.

Provides access to 100+ cloud services, including generative AI.

Enables migration, modernisation, and innovation on a fully managed infrastructure.



3.

Delivers a fully integrated cloud experience for IaaS, PaaS, and SaaS.

Offers industry-leading price-performance.

Ensures the highest levels of security.

4.

Gain increased control over Oracle workload placement.

Run Oracle SaaS products, including ERP, Financials, HCM, SCM, EPM, and CRM with best-in-class pre-integrated applications.

Leverage an integrated cloud solution supporting on-premises IaaS, PaaS, and SaaS.

Simplify deployment with preconfigured modular racks for easy setup.

Figure 6: Oracle Private Cloud - Benefits

5.3.1.1 Oracle Private Cloud - A Complete Cloud at Core42 Data Centre

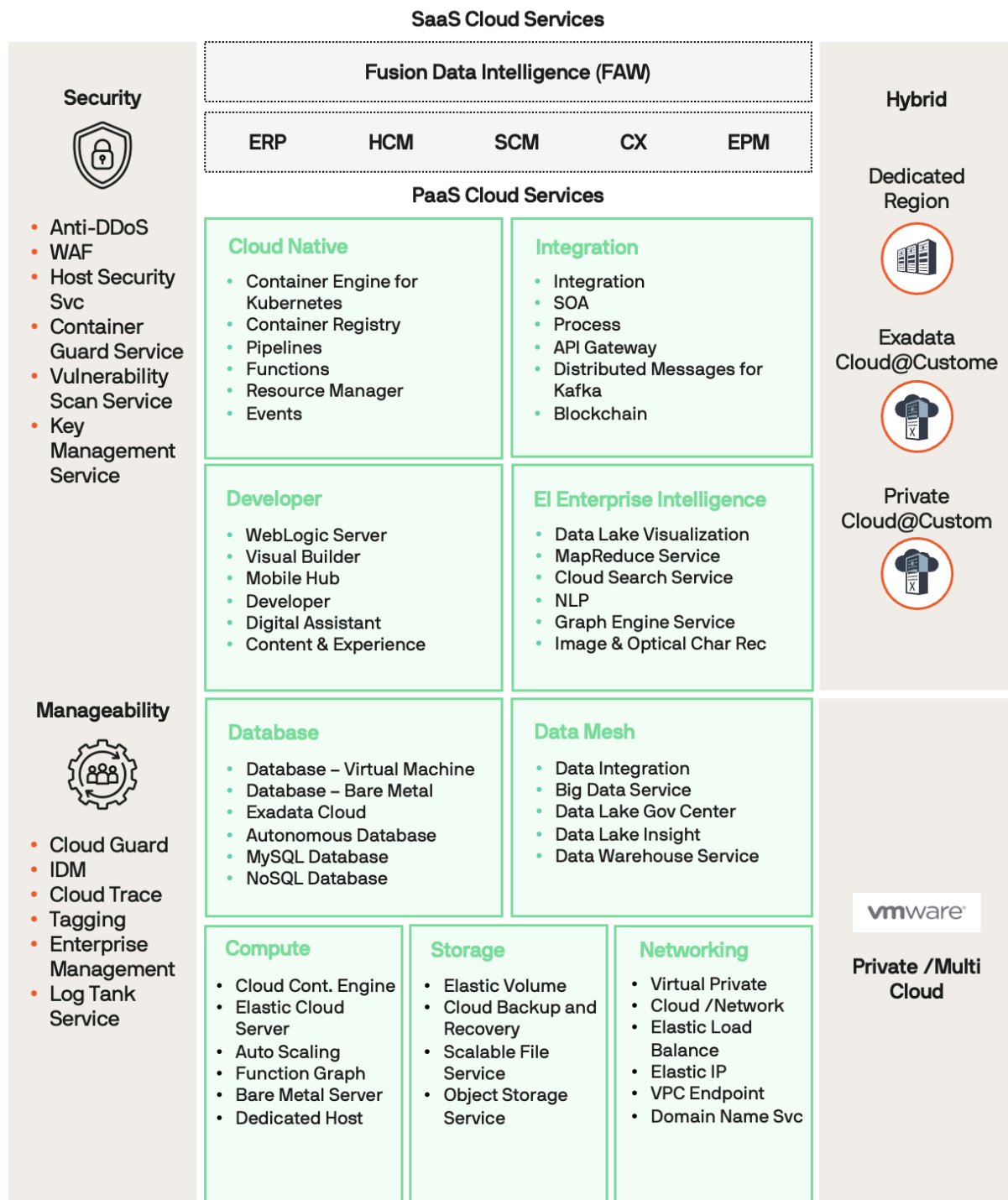


Figure 7: Oracle Private Cloud - A Complete Cloud at Core42 Data Centre

Oracle Private Cloud, delivered through Core42, seamlessly integrates Software-as-a-Service (SaaS) and Platform-as-a-Service offerings. Access to Oracle's comprehensive suite of cloud services within the Core42 data centre ensures unparalleled scalability, security, and reliability to drive digital transformation while allowing DCT to maintain complete control over DCT's data.

- **Core Infrastructure Capabilities** include compute for virtual machines, bare metal, containers, Kubernetes, storage, networking, load balancing, graphical processing (GPU), and much more.
- **Data Management Capabilities** include a wide range of services for data and AI, data science, Oracle database and MySQL, applications, and analytics.
- **Developer Services** include those for low code, a widely adopted APEX low-code development platform, and tools for application development and APIs. Oracle was one of the first vendors to adopt Terraform for Infrastructure-as-Code broadly.
- **Governance and Administration capabilities** include identity management, various services for cloud security, and observability for the entire stack of infrastructure and applications.

5.3.1.2 Core42 Oracle Private Cloud Licensing

Core42 proposes two options for the Core42 Oracle Private Cloud BOQ:



Figure 8: Licensing Options

5.3.1.2.1 Bring Your Own Licenses Subscription

Oracle BYOL allows DCT to apply licenses they currently own for on-premises software for equivalent, highly automated Oracle PaaS, helping DCT reach business value faster.

The Oracle BYOL to PaaS includes **compute**, **compute support**, and **automation** components and constitutes the following features:

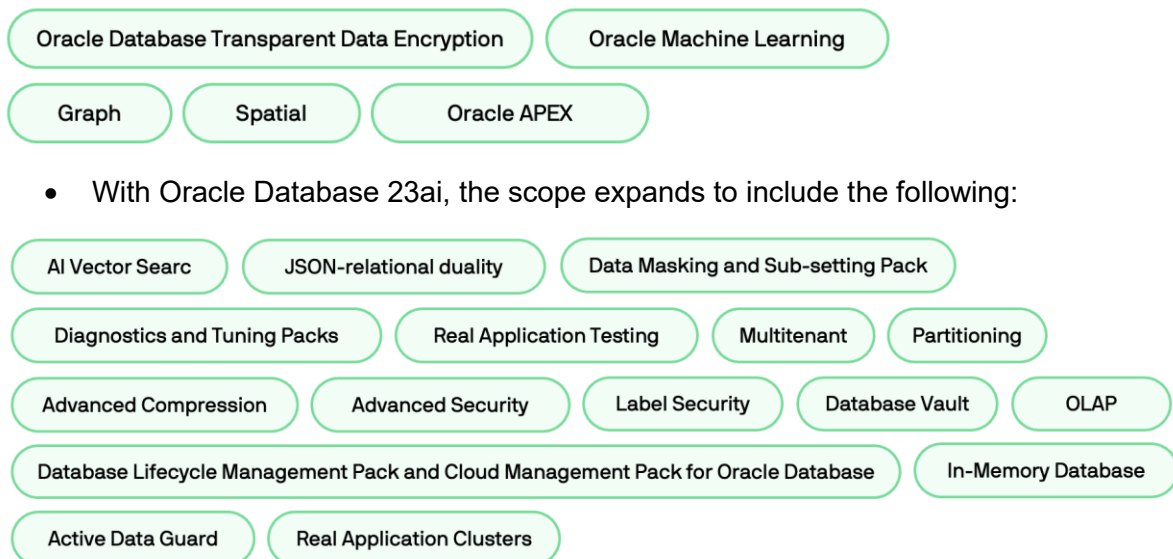
- Customers bring their on-premises license entitlement and get license support via their existing on-premises support contract. Customers can move to the cloud at a lower cost as they leverage their existing on-premises license entitlement.
- Customers must continue to pay their annual support fees for on-premises licenses they use for BYOL to PaaS.
- When customers bring an Oracle Database Enterprise Edition license entitlement to Oracle PaaS, they are granted the rights to use diagnostics pack, tuning pack, data masking and sub-setting pack, Oracle Database Transparent Data Encryption, Oracle machine learning, Graph, Spatial, and Oracle APEX. With Oracle Database 23ai, the scope expands to include AI Vector Search, JSON-relational duality, and real application testing without having on-premises license entitlements for those database options.
- If DCT elects to run Oracle Database Enterprise Edition and any of the eligible options/management packs listed below, then DCT's BYOL requirements are as follows:
 - Oracle Database Enterprise Edition plus a license for each database option/management pack that is elected to run in DCT's cloud environment.

- Eligible options include active data guard, advanced analytics, advanced compression, advanced security database InMemory, database lifecycle management pack, database vault, label security, multitenant, OLAP, partitioning, real application clusters, and spatial and graph.
- DCT may use the following options in the BYOL cloud service for use only with the BYOL cloud service without a supported license: data masking and sub-setting pack, diagnostics pack, tuning pack, real application testing, and the DDL logging functionality of the database lifecycle management pack.
- DCT may use the transparent database encryption feature in the BYOL cloud service only with the BYOL cloud service without a supported license of the **advanced security** database option.

5.3.1.2.2 License Included Subscription

The **License Included** option of Oracle's PaaS offering offers license-included PaaS. It includes the following:

- Compute and compute support.
- Automation.
- License entitlement and license support.
- Oracle Base Database Enterprise Edition - extreme performance considered in this option includes the following products:



- With Oracle Database 23ai, the scope expands to include the following:

Core42's Oracle Private Cloud offers a range of powerful cloud services, including:

- Oracle Autonomous Database for simplified database management.
- Oracle Cloud Infrastructure for scalable and secure infrastructure.
- Oracle Cloud Applications for streamlined business processes.
- Oracle Analytics Cloud for data-driven insights.
- Oracle Integration Cloud for seamless integration.

- Oracle Security Cloud for enhanced security and compliance.
- Oracle AI and machine learning for intelligent automation.

These services provide enhanced security and control over data and applications, improved scalability, flexibility to meet changing business needs, and streamlined business processes. By leveraging Core42's Oracle Private Cloud, DCT can drive significant business value, improve operational efficiency, and enhance Customer experience.

5.3.1.3 Cloud Platform Operation Support

- Core42 will support Oracle Private Cloud services in conjunction with Oracle support. DCT can raise a service ticket directly with Core42.
- Core42 will take ownership of resolving any data centre and network issues identified and escalated by Oracle support.
- DCT's IT support team will troubleshoot and diagnose any external network-related issues affecting DCT connectivity and integrations.
- Seamless collaboration will be maintained between Oracle, Core42, and DCT's IT team to ensure timely and efficient resolution of technical issues across the Oracle Private Cloud environment.

5.3.1.4 Service Level Agreements (SLA)

Enterprises demand more than just availability from their cloud infrastructure. Mission-critical workloads also require consistent performance and the ability to manage, monitor, and modify resources running in the cloud at any time. Oracle Private Cloud offers end-to-end SLAs covering service performance, availability, and manageability.

- **Availability:** Rest assured that DCT's cloud workloads are continuously operating, with Core42's commitments to uptime and connectivity backed by a full back-to-back agreement with Oracle.
- **Manageability:** The elasticity and configurability of infrastructure are part of why people move applications to the cloud. DCT's services need to be manageable all the time to deliver this benefit. Core42 with Oracle provides manageability SLAs to ensure DCT's ability to manage, monitor, and modify resources.
- **Performance:** It is not enough for DCT's IaaS resources to be merely accessible. They should consistently perform the way DCT expect them to. Core42, working with Oracle, is the first cloud vendor partnership to guarantee performance so that DCT can rely on DCT's infrastructure for enterprise applications.

Note: The SOW and a detailed SLA/SLO for each of the cloud services will be shared as per the cloud hosting and delivery policies.

5.3.1.5 Oracle Private Cloud Services Capabilities

The proposed solution leverages a secure, high-performance, and scalable environment designed to provide the flexibility and reliability required for seamless integration across DCT's infrastructure. It incorporates powerful compute options, flexible storage, and a robust networking framework, ensuring operational efficiency and system reliability.

Capabilities of Core42's Oracle Private Cloud include:

- **Scalability, Flexibility, and Control:** Core42's Oracle Private Cloud combines Oracle's public cloud services within a private and secure environment, striking the perfect balance of scalability, flexibility, and control.
- **Comprehensive Integrated Cloud Environment:** This solution provides a comprehensive, integrated cloud environment that addresses data residency and security concerns while empowering growth and technological advancement.
- **Powered by Oracle Cloud Infrastructure:** The Oracle Private Cloud is powered by Oracle Cloud Infrastructure, offering a dedicated region that brings the full power of cloud services as a devoted cloud.
- **Fully Managed Cloud Region:** Core42's fully managed cloud region offers a flexible and scalable dedicated infrastructure optimised for diverse environments, allowing Customers to deploy infrastructure dedicated to their enterprise in the location of their choice.
- **Public Cloud Benefits with Security and Compliance:** The solution provides the benefits of the public cloud while addressing strict security, regulatory, and data residency requirements in two regions across Abu Dhabi.
- **Access to Over 100 Cloud Services:** Customers can access over 100 cloud services to migrate, modernise, and innovate on a fully managed infrastructure.
- **Integrated Cloud Experience:** The solution offers a fully integrated cloud experience for IaaS, PaaS, and SaaS with industry-leading price-performance and the highest levels of security.
- **Increased Control Over Oracle Workloads:** The dedicated region allows Customers to achieve increased control over where they run Oracle workloads.

These capabilities make Core42's Oracle Private Cloud an ideal choice for modernising infrastructure and supporting sustained success.

5.3.1.6 Oracle Private Cloud - Core Elements

Oracle Private Cloud's infrastructure-as-a-service (IaaS) features give businesses a strong and flexible foundation to support their IT needs. Powered by Oracle Dedicated Region Cloud@Customer, it brings high-performance computing, adaptable storage, and secure networking right to DCT's data centre. Designed to meet the unique challenges of modern enterprises, it ensures everything runs smoothly with reduced latency and seamless integration, all while giving DCT complete control over DCT's infrastructure-helping DCT's business work smarter and innovate faster.

IaaS services include the following core elements:

To be updated based on the offering

- Compute
- Block, object, and file storage
- Database services
- Network services
- PaaS services for developer tools

- Oracle Private Cloud security
- Platform security
- Physical security
- Operational security
- Fault-tolerant infrastructure for high availability
- Core42 Oracle Private Cloud Licensing
- Cloud Platform Operation Support
- Service Level Agreements

6 Reference Architectures

6.1 Solution Architecture Overview

The following subsections present the reference architectures underpinning the proposed solution, covering the high-level design, disaster recovery framework, and security posture to collectively ensure a resilient, secure, and enterprise-ready deployment.

6.1.1 High-Level Architecture

The following diagram presents the high-level architecture.

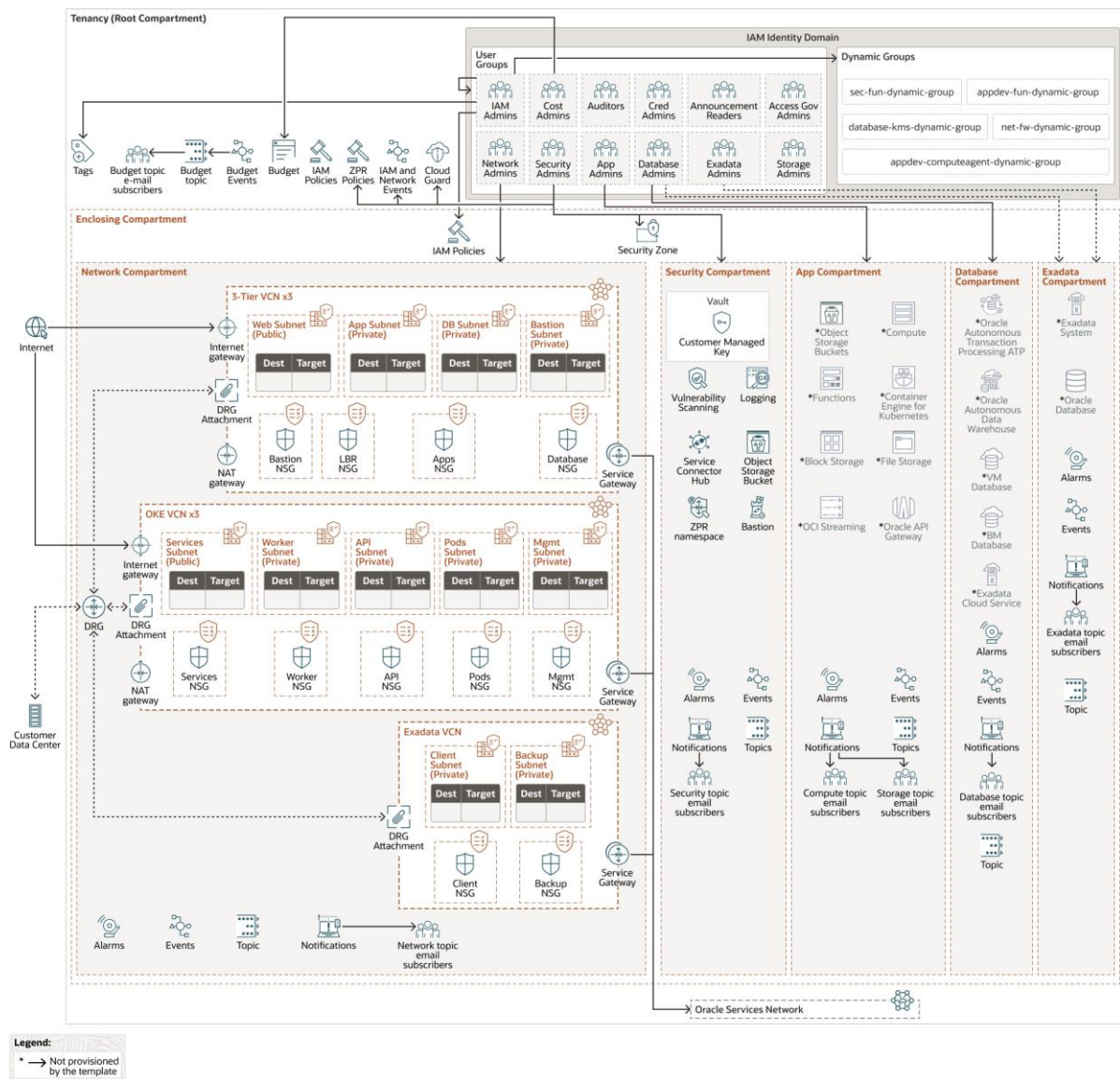


Figure 9: High-Level Architecture

Tenancy

A tenancy is a secure and isolated partition that Oracle sets up within Oracle Cloud when DCT sign up for Oracle Cloud Infrastructure. DCT can create, organise, and administer your resources in Oracle Cloud within your tenancy. A tenancy is synonymous with a company or

organisation. Usually, a company will have a single tenancy and reflect its organisational structure within that tenancy. A single tenancy is usually associated with a single subscription, and a single subscription usually only has one tenancy.

Identity domain

An identity domain is a container for managing users and roles, federating and provisioning users, securing application integration through single sign-on (SSO) configuration, and SAML/OAuth-based identity provider administration. It represents a user population in Oracle Cloud Infrastructure and its associated configurations and security settings (such as MFA).

Policies

An Oracle Cloud Infrastructure Identity and Access Management policy specifies who can access which resources, and how. Access is granted at the group and compartment level, which means DCT can write a policy that gives a group a specific type of access within a specific compartment, or to the tenancy.

Compartments

Compartments are cross-regional logical partitions within an Oracle Cloud Infrastructure tenancy. Use compartments to organise, control access, and set usage quotas for your Oracle Cloud resources. In a given compartment, DCT define policies that control access and set privileges for resources.

The resources in this landing zone template are provisioned in the following compartments:

- A recommended enclosing compartment containing all compartments listed below.
- A Network compartment for all the networking resources, including the required network gateways.
- A Security compartment for the logging, key management, and notifications resources.
- An App compartment for the application-related services, including compute, storage, functions, streams, Kubernetes nodes, API gateway, and so on.
- A Database compartment for all database resources.
- An optional compartment for Oracle Exadata Database Service infrastructure.

The greyed-out icons in the diagram indicate services that are not provisioned by the template.

This compartment design reflects a basic functional structure observed across different organisations, where IT responsibilities are typically separated among networking, security, application development, and database administrators.

Virtual cloud network (VCN) and subnets

A VCN is a customizable, software-defined network that DCT set up in an Oracle Cloud Infrastructure region. Like traditional data center networks, VCNs give DCT control over your network environment. A VCN can have multiple non-overlapping CIDR blocks that DCT can change after DCT create the VCN. DCT can segment a VCN into subnets, which can be

scoped to a region or to an availability domain. Each subnet consists of a contiguous range of addresses that don't overlap with the other subnets in the VCN. DCT can change the size of a subnet after creation. A subnet can be public or private.

The template can deploy VCNs for different workload types, including three-tier VCNs for typical three-tier web-based applications, OCI Kubernetes Engine applications and Oracle Exadata Database Service.

Internet gateway

An internet gateway allows traffic between the public subnets in a VCN and the public internet.

Dynamic routing gateway (DRG)

The DRG is a virtual router that provides a path for private network traffic between on-premises networks and VCNs and can also be used to route traffic between VCNs in the same region or across regions.

NAT gateway

A NAT gateway enables private resources in a VCN to access hosts on the internet, without exposing those resources to incoming internet connections.

Service gateway

The service gateway provides access from a VCN to other services, such as Oracle Cloud Infrastructure Object Storage. The traffic from the VCN to the Oracle service travels over the Oracle network fabric and does not traverse the internet.

Oracle services network

The Oracle Services Network (OSN) is a conceptual network in Oracle Cloud Infrastructure that is reserved for Oracle services. These services have public IP addresses that DCT can reach over the internet. Hosts outside Oracle Cloud can access the OSN privately by using Oracle Cloud Infrastructure FastConnect or VPN Connect. Hosts in your VCNs can access the OSN privately through a service gateway.

Network security groups (NSGs)

NSGs act as virtual firewalls for your cloud resources. With the zero-trust security model of Oracle Cloud Infrastructure DCT control the network traffic inside a VCN. An NSG consists of a set of ingress and egress security rules that apply to only a specified set of VNICs in a single VCN.

Events

Oracle Cloud Infrastructure services emit events, which are structured messages that describe the changes in resources. Events are emitted for create, read, update, or delete (CRUD) operations, resource lifecycle state changes, and system events that affect cloud resources.

Notifications

OCI Notifications broadcasts messages to distributed components by using a low latency publish-subscribe pattern, delivering secure, highly reliable, durable messages for applications hosted on Oracle Cloud Infrastructure.

Vault

Oracle Cloud Infrastructure Vault enables DCT to centrally manage the encryption keys that protect your data and the secret credentials that DCT use to secure access to your resources in the cloud. DCT can use the Vault service to create and manage vaults, keys, and secrets.

Logs

Oracle Cloud Infrastructure Logging is a highly-scalable and fully-managed service that provides access to the following types of logs from your resources in the cloud:

- Audit logs: Logs related to events produced by OCI Audit.
- Service logs: Logs published by individual services such as OCI API Gateway, OCI Events, OCI Functions, OCI Load Balancing, OCI Object Storage, and VCN flow logs.
- Custom logs: Logs that contain diagnostic information from custom applications, other cloud providers, or an on-premises environment.

Service connectors

Oracle Cloud Infrastructure Connector Hub is a cloud message bus platform that orchestrates data movement between services in OCI. DCT can use service connectors to move data from a source service to a target service. Service connectors also enable DCT to optionally specify a task (such as a function) to perform on the data before it is delivered to the target service.

DCT can use Oracle Cloud Infrastructure Service Connector Hub to quickly build a logging aggregation framework for SIEM systems.

Cloud Guard

Oracle Cloud Guard helps DCT achieve and maintain a strong security posture in Oracle Cloud by monitoring the tenancy for configuration settings and actions on resources that could pose a security problem.

DCT can use Oracle Cloud Guard to monitor and maintain the security of your resources in Oracle Cloud Infrastructure. Cloud Guard uses *detector recipes* that DCT can define to examine your resources for security weaknesses and to monitor operators and users for certain risky activities. When any misconfiguration or insecure activity is detected, Cloud Guard recommends corrective actions and assists with taking those actions, based on *responder recipes* that DCT can define.

Security zone

A security zone is associated with one or more compartments and a security zone recipe. When DCT create and update resources in a security zone, Oracle Cloud Infrastructure (OCI) validates these operations against the list of policies that are defined in the security zone recipe. If any security zone policy is violated, then the operation is denied.

Security zones ensure that your OCI resources comply with your security policies, including Oracle Cloud Infrastructure Compute, Oracle Cloud Infrastructure Networking, Oracle Cloud Infrastructure Object Storage, Oracle Cloud Infrastructure Block Volumes, and Database resources.

Vulnerability Scanning Service

Oracle Cloud Infrastructure Vulnerability Scanning Service helps improve the security posture in Oracle Cloud by routinely checking ports and hosts for potential vulnerabilities. The service generates reports with metrics and details about these vulnerabilities.

Zero Trust Packet Routing

Oracle Cloud Infrastructure Zero Trust Packet Routing (ZPR) prevents unauthorised data access by managing the network security policy separately from the network architecture. ZPR uses a user-friendly, intent-based policy language to define allowed access pathways for data. Any traffic patterns not explicitly defined by policy cannot traverse the network, which simplifies data protection and prevents data exfiltration.

Bastion service

Oracle Cloud Infrastructure Bastion service provides restricted access from specific IP addresses to target OCI resources that do not have public endpoints using Identity-based, audited and time-bound Secure Shell (SSH) sessions.

Object Storage

OCI Object Storage provides quick access to large amounts of structured and unstructured data of any content type, including database backups, analytic data, and rich content such as images and videos. DCT can safely and securely store data directly from the internet or from within the cloud platform. DCT can scale storage without experiencing any degradation in performance or service reliability.

6.1.2 Disaster Recovery Architecture

The following diagram presents the high-level Disaster Recovery architecture.

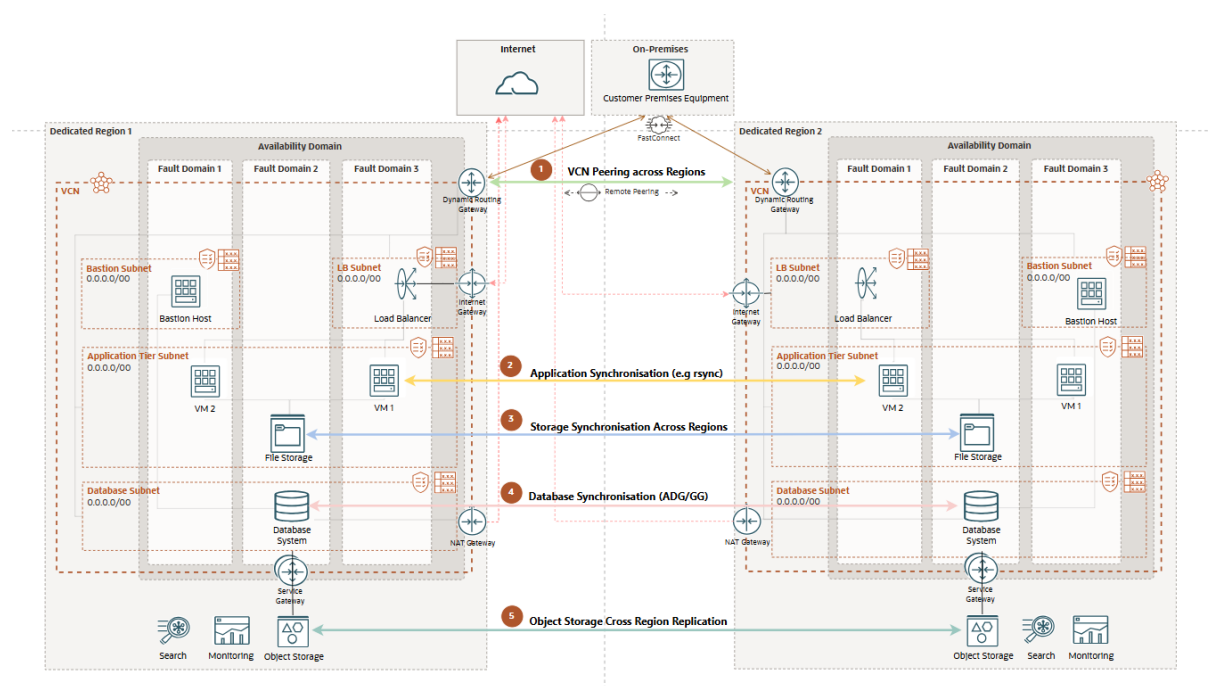


Figure 10: Disaster Recovery Architecture

6.1.3 Security Architecture Posture

The following diagram presents the high-level Security architecture posture.

OCI MTLZ - SECURITY VIEW - SECURITY POSTURE															BACK TO INDEX		
TENANCY SECURITY															WORKLOAD SECURITY		
COMPARTMENTS	IDENTITY DOMAINS	GROUPS	POLICIES	CLOUD GUARD	SECURITY ZONES	QUOTAS	NETWORK SECURITY	EVENTS	LOGGING	ALARMS	NOTIFICATIONS	BASTION	VAULTS	VULNERABILITY SCANNING			
	Aggregates Identities and Permissions	Provides Users, Aggregate into Roles	Defines access level from Groups to Compartments		Enables operations comply with security standards (data protection) (P)	Enables Service Limits by Compartments	Enables traffic control and controlled access for resources	Enables metadata response to resource state change (data)	Enables metadata response to resource tagging (metadata response) (data)	Enables metadata response to thresholding events on resource (data)	Enables the notification of security events of systems						
ROOT	DEFAULT	BREAK GLASS	YES	YES	RECIPE 01 or RECIPE 02	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
LANDING ZONE ENVIRONMENT (N)	NO	YES	YES	N/A	NO (OPTIONAL)	YES	NO	YES	NO	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
SHARED NETWORK (S)	NO	YES	YES	N/A	RECIPE 03	NO (OPTIONAL)	AREA 1	CONNECTIVITY, NETWORK, SECURITY ZONES	SITE-TO-SITE VPN, FLUX LOSS FIREWALLS	SITE-TO-SITE FASTCONNECT	NO	NO	NO	NO	NO	NO	NO
SHARED SECURITY (S)	NO	YES	YES	N/A	NO	NO (OPTIONAL)	NO	SECURITY ZONES NOTIFICATIONS	NO (OPTIONAL)	NO (OPTIONAL)	YES	YES	YES	YES	YES	YES	YES
SHARED PLATFORMS (S)	NO	YES	YES	N/A	RECIPE 05	NO (OPTIONAL)	NO	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
DEV / WORKLOAD ENVIRONMENT (N)	NO	YES	YES	N/A	NO (OPTIONAL)	YES	NO	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
WE NETWORK (S)	NO	YES	YES	N/A	RECIPE 04	NO (OPTIONAL)	AREA 2 AREA 4	NETWORK SECURITY ZONES	VPN FLUX LOSS	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
WE SECURITY (S)	NO	YES	YES	N/A	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	YES	YES	YES	YES	YES	YES	YES
PLATFORMS (S)	NO	NO (OPTIONAL)	YES	N/A	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
PLATFORMS (N)	NO (OPTIONAL)	YES	YES	N/A	RECIPE 05	NO (OPTIONAL)	AREA 3	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
PROJECTS (S)	NO	NO (OPTIONAL)	YES	N/A	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
PROJECT (N)	NO (OPTIONAL)	YES	YES	N/A	RECIPE 05	NO (OPTIONAL)	AREA 3	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO
LAYER (N)	NO	NO (OPTIONAL)	NO (OPTIONAL)	N/A	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO (OPTIONAL)	NO (OPTIONAL)	NO (OPTIONAL)	NO	NO	NO	NO	NO	NO	NO

Figure 11: Security Architecture Posture

7 Assumptions, Dependencies, and Out of Scope

7.1 Assumptions

This proposal relies on key assumptions essential for the project's successful delivery. Any modifications or deviations from these assumptions could affect the project's scope, schedule, and budget.

- DCT will bring necessary license other than included in the BOQ.
- No PS included in this proposal, Core42 will only provision a new Tenancy for DRCC admin team.

7.2 Out of Scope

The services outlined below are outside the current project scope. Any further requests concerning these activities need a separate agreement and could lead to extra costs and delays.

- Connectivity – Any communication link.
- Any License other than mentioned in the BOQ.
- Professional Service for Cloud resource provisioning and configuration.

8 Solution Deep Dive

This section provides detailed information on various topics from the previous sections.

8.1 Oracle Private Cloud - Core Elements

8.1.1 Compute (IaaS)

The compute in Oracle Private Cloud allows DCT to provision and manage compute hosts, known as instances, to meet DCT's compute and application requirements.

Instance Types

- **Bare Metal:** Dedicated physical server access for the highest performance and strong isolation.
- **Virtual Machine (VM):** An Independent computing environment that runs on top of physical bare metal hardware, ideal for applications that don't require the entire physical machine's resources.

Full Range of Compute Options

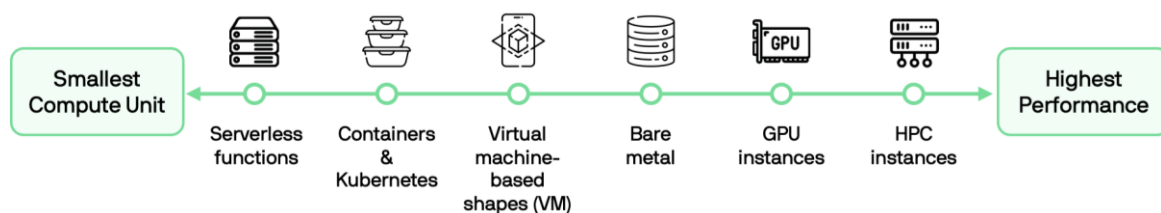


Figure 12: Full Range of Compute Options

Compute Shapes

Compute shapes determine the resources allocated to an instance. Shapes are available with AMD, Intel, and ARM-based processors.

Shape Options:

- **Standard Shapes:** General-purpose workloads, balanced cores, memory, and network resources.
- **Dense I/O Shapes:** Large databases, big data workloads, and high-performance local storage.
- **GPU Shapes:** Hardware-accelerated workloads, Intel or AMD CPUs, and NVIDIA graphics processors.
- **High Performance Computing (HPC):** High-frequency processor cores, bare metal HPC, and optimised shapes.
- **Optimised Shapes:** Flexible virtual machine and bare metal shapes for multiple workload types, including HPC and memory-optimised use cases.

8.1.2 Block, Object, and File Storage (IaaS)

Oracle Private Cloud provides Customers with scalable, high-performance computing and low-cost cloud storage options. This addresses key storage workload requirements and use

cases through on-demand local, object, file, block, and archive storage. Customers can use the data transfer service to move their data safely and securely to the cloud.

High Performance Storage

Oracle Private Cloud provides flexible and reliable storage options to fit DCT's needs. With block volumes, DCT can easily manage and move storage like a regular hard drive. Object storage lets DCT securely store and scale unstructured data like images or analytics with frequent or infrequent access options. For shared file systems, file storage offers a secure, scalable solution accessible anywhere in DCT's network.



Figure 13: High Performance Storage of Block, Object, and File Storage (IaaS)

Block Volumes (Storage)

The Block Volumes service lets DCT dynamically provision and manage **Block Storage Volumes**. Customers can create, attach, connect, and move volumes and change volume performance as needed to meet their storage, performance, and application requirements. After DCT attach and connect a volume to an instance, DCT can use it like a regular hard drive. DCT can also disconnect a volume and attach it to another instance without losing data.

Object Storage

Object Storage is a high-performance, internet-scale storage platform that offers reliable and cost-efficient data durability for storing unlimited amounts of unstructured data, including analytic data, images, and videos. The service provides secure and direct access to data from the internet or within the cloud platform, with multiple management interfaces for easy scalability. Object storage offers three storage tiers:

- Standard (default, "hot" storage for frequently accessed data)
- Archive (for infrequently accessed data)
- Infrequent access (for data requiring less frequent access) allows users to choose the optimal tier for their data storage needs.

OCI File Storage

File Storage provides a durable, scalable, and secure enterprise-grade network file system that can be accessed from any instance in a Virtual Cloud Network (VCN) or outside the VCN using various connectivity options. The service is designed to meet the needs of a wide range of use cases, including general-purpose file storage, big data and analytics, lift and shift of enterprise applications, databases and transactional applications, backups and disaster recovery, and microservices and Docker, allowing users to easily manage and scale their file systems to meet their specific needs.

8.1.3 Database Services by Oracle Autonomous Database

Oracle Autonomous Database (PaaS) uses machine learning to automate database tuning, security, backups, updates, and other routine management tasks traditionally performed by

DBAs. Unlike a conventional database, Oracle Autonomous Database performs all these tasks and more without human intervention.

Benefits

- Maximum database uptime, performance, and security—including automatic patches and fixes.
- Elimination of manual, error-prone management tasks through automation.
- Reduced costs and improved productivity by automating routine tasks.
- With Oracle Autonomous Database, DCT can bring AI to DCT's data.
 - Get quick insights from DCT's data via contextual conversations in natural language, without complex manual operations.
 - Use Large Language Models (LLMs) with built-in AI Vector Search and DCT's proprietary data to get more accurate answers. There's no need to duplicate data to a separate vector database.
 - Rapidly build and deploy machine learning models in-database at scale—even without coding.

8.1.4 Network Services

FastConnect

- FastConnect is a dedicated, private connection between OCI and DCT's environment, with port speeds from 1G to 400G and no per-byte charge for data movement. The service is Oracle-managed (PaaS) and provides multiple configuration options for Customers.
- Use FastConnect to extend DCT's existing infrastructure into a virtual cloud network (VCN) in Oracle Private Cloud, ideal for hybrid cloud or lift-and-shift scenarios. This lets DCT communicate with private IPv4 addresses (RFC 1918) across the connection.

8.1.5 PaaS Services for Developer Tools

OCI Resource Manager

Resource Manager is a PaaS service that automates deployment and operations for all OCI resources using open-source Terraform software, a popular industry standard for Infrastructure-as-Code (IaC). The service comes with a pre-installed Terraform host that includes a range of developer tools, including Ansible and Docker, allowing DevOps engineers to develop and deploy their infrastructure efficiently and effectively.

Oracle Visual Builder Studio

Oracle Visual Builder Studio is a robust application development PaaS that helps DCT's team effectively plan and manage DCT's work throughout all stages of the AppDev lifecycle: design, build, test, and deploy. It makes it easy for DCT's team to develop the necessary artefacts.

- Oracle Cloud applications developers, who need to extend their applications with business-specific customisations.

- Low-code developers, who want to create applications using a visual designer.
- Experienced programmers who want to modify the source code for applications created by others or to develop bespoke apps using the web programming language of their choice.

Oracle APEX

Oracle APEX is a fully managed PaaS, low-code application development platform for building and deploying modern cloud, mobile, and data-driven applications in Oracle Cloud. Business users and application developers can create enterprise applications 20X faster than coding-without learning complex web technologies.

APEX provides an easy-to-use browser-based environment to load data, manage database objects, develop REST interfaces, and build applications that look and run great on desktop and mobile devices. DCT can use APEX to develop solutions that perform a wide variety of tasks, including the following:

- Import spreadsheets and develop a single source of truth in minutes.
- Create compelling data visualisations against DCT's existing data.
- Deploy productivity applications to solve a business need elegantly.
- Manage mission-critical data.

8.1.6 Oracle Private Cloud Security

Oracle Private Cloud is designed to secure the platform through isolated network virtualisation, highly secure firmware installation, a controlled physical network, and network segmentation.

Platform Security

The physical network in Oracle Private Cloud is designed with security at its core, using segmentation and strict access controls to protect DCT environments and minimise risks. Enforcing access policies and closely monitoring service areas ensures a secure and transparent network experience.

Isolated Network Virtualisation

Oracle Private Cloud features isolated network virtualisation, significantly reducing the risk associated with traditional virtualisation environments. By decoupling network virtualisation from the hypervisor, OCI moves cloud control to a highly customised hardware and software layer, limiting the attack surface and containing potential threats to a single host. This design prevents malicious actors from accessing other hosts in the cloud infrastructure, even in the event of a successful VM escape attack. As a result, all tenants benefit from reduced risk and enhanced security, with isolated network virtualisation implemented in every data centre across all regions.

Physical Network and Network Segmentation

The physical network architecture of Oracle Private Cloud adds a layer of defence to the isolated network virtualisation, further isolating DCT tenancies and limiting the risk of threat proliferation. The physical network components, including racks, routers, and switches, are designed to enforce access control lists (ACLs) that prevent IP spoofing and ensure that only

authorised traffic is allowed. The physical layer is a simple, flat network connected to virtual ports on the virtual cloud network (VCN), reducing complexity and increasing visibility of attempts to circumvent security measures.

Data Residency	Core42 Oracle Signature Fusion Cloud by Oracle Cloud Infrastructure (OCI) will have its own realm that does not federate or replicate with Oracle public cloud regions.	Ready
Data Security	- Data at rest and in transit always encrypted (minimum AES 256 and TLS 1.2 respectively) - End-user data can always be encrypted with end-user- controlled encryption keys. - Oracle has no access or control over cryptographic keys backed by external HSM in customer premises.	Ready
Access Control	- Customer tenancy will be accessed only by Customer and neither Oracle nor Core42 operation team have access to the entire tenancy including data. - Only Telemetry, Service availability and Performance metrics shall be shared with Oracle operations team for administrative purposes (consist of UID (Unique Identifiers) and hashed at SHA 256) and does NOT contain PII.	Ready
Regulatory Compliance	Solution is compliant with UAE regulations including UAE IA, (formerly NES), ADISS and DESC.	5/7 Available
Security Operations	Oracle will provide Core42 with additional information about the security operations of their dedicated regions, through the following: - OCI service patch activity aggregated summary reports. - Quarterly review of security alerts: detection, response, and status within Core42 Oracle National Cloud. - Upon request, Oracle will provide mutually agreed upon Secure Shell (SSH) audit logs for activities performed by OCI operators.	Ready
Physical Security	Local Core42 personnel control data center physical security including physical access.	Ready
Disaster Recovery (DR)	Two data regions provide Disaster Recovery (DR) entirely in the UAE.	Ready

Figure 14: Physical Network Architecture of Oracle Private Cloud

Network Segmentation and Access Control

The physical network is segmented into enclaves with unique communications profiles, controlling access into and out of these enclaves. Compute hosts are power cycled by an Integrated Lights Out Manager (ILOM), and direct communication with other hosts is prohibited. Access to the services enclave, where core Cloud services are provisioned, is controlled, monitored, and policy-driven, with explicit user privileges granted by authorised persons. This access is subject to regular auditing and review, ensuring the security and integrity of the network.

Physical Security

Oracle Private Cloud data centres are designed for resilience, meeting Tier 3 standards with strong physical security and redundancy to handle any threats. Access to systems is tightly controlled using least-privilege principles, multi-factor authentication, secure VPNs, and regular audits to keep DCT's data safe.

Physical Security and Secure Connectivity

Oracle Private Cloud data centres undergo an extensive risk-evaluation process and align with Uptime Institute and Telecommunications Industry Association (TIA) ANSI/TIA-942-A Tier 3 or Tier 4 standards. Data centres with redundant power sources, generator backups, and fire suppression systems are designed to withstand environmental threats, power outages, and other risks. Physical security measures include perimeter barriers, security guards, cameras, and two-factor access control. Server rooms are isolated with physical obstacles, and access is granted only to authorised personnel.

Operational Security

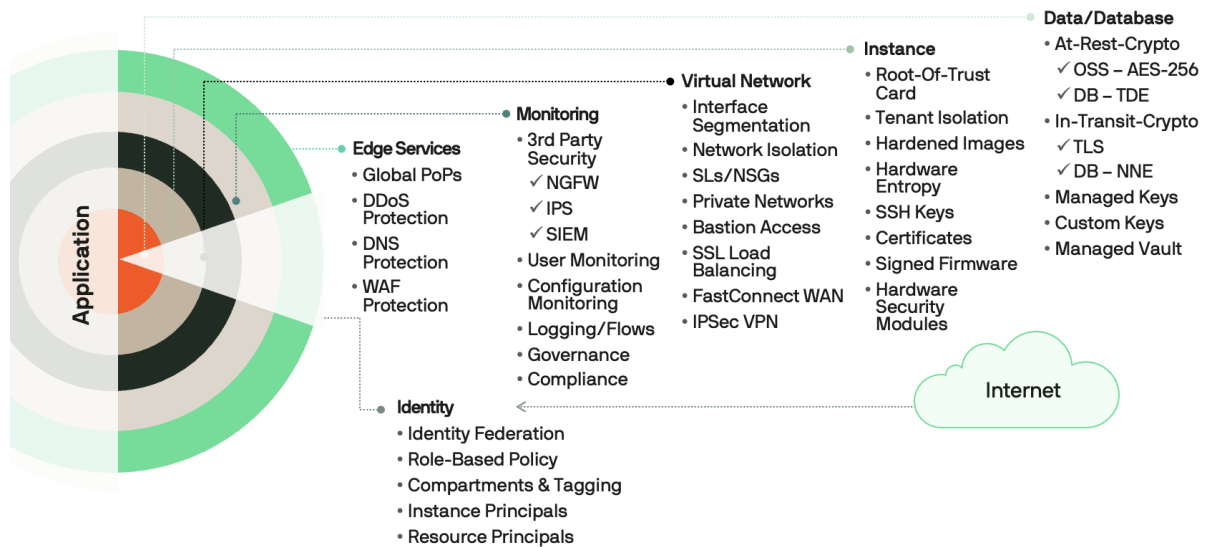


Figure 15: Operational Security

Defensive Security

Daily attacks occur against networking and computing infrastructure in all computing environments. A dedicated team of defensive experts and analysts must monitor and respond to these events. They work proactively and continuously to spot potential threats and shut down exploit paths. When incidents are detected, they promptly remediate them using modern security operations methodologies and DevSecOps-enabled configuration and tooling.

Offensive Security

The offensive security team verifies that it meets security benchmarks and best practices. This team works to understand and emulate attackers' methods, including sophisticated bad actors and nation-states. This work involves research, penetration testing, and simulating advanced threats against Oracle hardware and software. The offensive security team's work informs secure development, architecture, and defensive capabilities.

Security Assurance

The SFC platform is built and implemented with robust security plans that adhere to stringent security standards and align with Oracle's and industry-wide best practices. To ensure the cloud platform's security, the security assurance group collaborates closely with service teams and security stakeholders across Oracle to design, implement, and maintain security controls, technologies, processes, and guidelines for teams utilising Oracle Cloud Infrastructure.

Incident Response

Implemented security mechanisms that combine strict processes and experienced cybersecurity experts with a defence-in-depth cloud architecture.

Least-Privilege Access and Multiple Authentication Layers

SFC uses the principle of least-privilege access to grant access to production systems, with periodic reviews and revocation of unnecessary permissions. Multifactor authentication (MFA) is required to access production systems, with logs kept for security analysis. Multiple

layers of advanced access control are used to meter access to network devices and servers, including compulsory virtual private network (VPN) connectivity with high password diversity and Universal 2nd Factor (U2F) authentication. All administrative access is logged, and access permissions are audited for least-privilege to prevent attackers from accessing the administrative network with weak or breached passwords.

8.1.7 Fault-Tolerant Infrastructure for High Availability

Fault tolerance is implemented in the service architecture and in data storage. Services and data span racks of hardware, which themselves include multiple layers of redundancy at the node, server, and hardware component levels. Connectivity and edge services link each region with other regions, peering networks, and DCT data centres.

- Each availability domain includes three fault domains.
- Fault domains enable DCT to distribute DCT's resources so that they don't depend on the same physical hardware within the OCI region.
- A Fault Domain is a grouping of hardware and infrastructure within an Availability Domain that provides anti-affinity (logical data centre).
- As a result, hardware failures or maintenance events that affect one fault domain do not affect the resources in other fault domains.

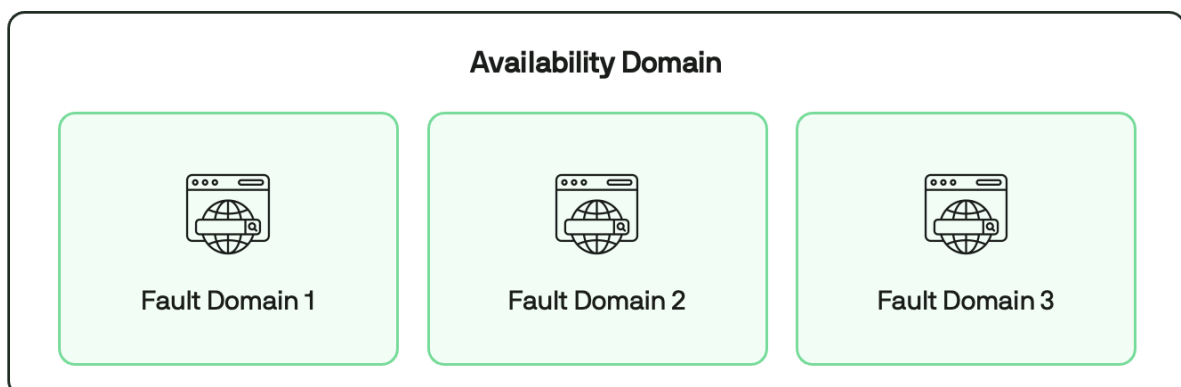


Figure 16: Fault-Tolerant Infrastructure for High Availability

8.1.7.1 Core42 Oracle Private Cloud Licensing

Core42 proposes two options for the Core42 Oracle Private Cloud BOQ:



Figure 17: Licensing Options

Bring Your Own Licenses Subscription

BYOL stands for "bring your own license." Oracle BYOL allows Customers to apply licenses they currently own for on-premises software to equivalent, highly automated Oracle PaaS, helping them reach business value faster.

The Oracle BYOL to PaaS includes **Compute**, **Compute Support**, and **Automation** components and constitutes the following features.

- Customers bring their on-premises license entitlement and get license support via their existing on-premises support contract. Customers can move to the cloud at a lower cost as they leverage their existing on-premises license entitlement.
- Customers must continue to pay their annual support fees for on-premises licenses they use for BYOL to PaaS.
- When a Customer brings an Oracle Database Enterprise Edition license entitlement to Oracle PaaS, they are granted the rights to use Diagnostics Pack, Tuning Pack, Data Masking and Sub-setting Pack, Oracle Database Transparent Data Encryption, Oracle Machine Learning, Graph, Spatial, and Oracle APEX. With Oracle Database 23ai, this expands to include AI Vector Search, JSON-relational duality and Real Application Testing without having on-premises license entitlements for those database options.
- If DCT elects to run Oracle Database Enterprise Edition and any of the eligible options/management packs listed below, then DCT's BYOL requirements are as follows:
 - Oracle Database Enterprise Edition plus a license for each database option/management pack DCT elect to run in DCT's Cloud environment. Eligible options include Active Data Guard, Advanced Analytics, Advanced Compression, Advanced Security Database InMemory, Database Lifecycle Management Pack, Database Vault, Label Security, Multitenant, OLAP, Partitioning, Real Application Clusters and Spatial and Graph.
- DCT may use the following options in the BYOL Cloud Service for use only with the BYOL Cloud Service without a supported license:
 - Data Masking and Sub-setting Pack, Diagnostics Pack, Tuning Pack, and Real Application Testing, and the DDL Logging functionality of the Database Lifecycle Management Pack.
 - Additionally, DCT may use the Transparent Database Encryption feature in the BYOL Cloud Service only with the BYOL Cloud Service without a supported license of the Advanced Security database option.

License Included Subscription

The License Included Option of Oracle's PaaS offering offers license-included PaaS, which includes the following.

- Compute and compute support
- Automation
- License entitlement and license support

Oracle Base Database Enterprise Edition - Extreme Performance, which is considered in this option, includes the following:

Oracle Database Transparent Data Encryption, Oracle Machine Learning, Graph, Spatial, and Oracle APEX. With Oracle Database 23ai, this expands to include AI Vector Search, JSON-relational duality, Data Masking and Sub-setting Pack, Diagnostics and Tuning Packs, and Real Application Testing, Multitenant, Partitioning, Advanced Compression, Advanced

Security, Label Security, Database Vault, OLAP, Database Lifecycle Management Pack and Cloud Management Pack for Oracle Database, In-Memory Database, Active Data Guard, Real Application Clusters.

Core42's Oracle Private Cloud offers a range of powerful cloud services, including:

- Oracle Autonomous Database for simplified database management.
- Oracle Cloud Infrastructure for scalable and secure infrastructure.
- Oracle Cloud Applications for streamlined business processes.
- Oracle Analytics Cloud for data-driven insights.
- Oracle Integration Cloud for seamless integration.
- Oracle Security Cloud for enhanced security and compliance.
- Oracle AI and Machine Learning for intelligent automation.

These services provide enhanced security and control over data and applications, improved scalability, flexibility to meet changing business needs, and streamlined business processes. By leveraging Core42's Oracle Private Cloud, DCT can drive significant business value, improve operational efficiency, and enhance DCT experience.

8.1.7.2 Cloud Platform Operation Support

- Core42 will provide support for Oracle Private Cloud services in conjunction with Oracle Support. DCT can raise a service ticket directly with Core42.
- Core42 will take ownership of resolving any data centre and network issues identified and escalated by Oracle Support.
- DCT's IT support team will troubleshoot and diagnose any external network-related issues affecting DCT connectivity and integrations.
- Seamless collaboration will be maintained between Oracle, Core42, and DCT's IT team to ensure timely and efficient resolution of technical issues across the Oracle Private Cloud environment.

8.1.7.3 Service Level Agreements (SLA)

Enterprises demand more than just availability from their cloud infrastructure. Mission-critical workloads also require consistent performance and the ability to manage, monitor, and modify resources running in the cloud at any time. Oracle Private Cloud offers end-to-end SLAs covering service performance, availability, and manageability.

- **Availability:** Rest assured that DCT's cloud workloads are continuously operating, with Core42's commitments to uptime and connectivity backed by a full back-to-back agreement with Oracle.
- **Manageability:** The elasticity and configurability of infrastructure is part of why people move applications to the cloud. DCT's services need to be manageable all the time to deliver this benefit. Core42 with Oracle provides manageability SLAs to ensure DCT's ability to manage, monitor, and modify resources.
- **Performance:** It is not enough for DCT's IaaS resources to be merely accessible. They should consistently perform the way DCT expect them to. Core42, working with

Oracle, is the first cloud vendor partnership to guarantee performance so that DCT can rely on DCT's infrastructure for enterprise applications.

Note: The SOW and a detailed SLA/SLO for each of the cloud services will be shared as per the Cloud Hosting and delivery policies.

9 Price Schedule

Below is the pricing for Oracle Tenant DRCC:

Service Name	Duration (Months)	Total Price AED
PaaS/IaaS	12 Months	499,394
Total Price AED (Excluding UAE VAT)		499,394

Table 3: Pricing

9.1 Pricing Assumptions

- All prices contained in this Proposal shall be adjusted annually on Contract Anniversary Date in accordance with the percentage change in the latest Annual Consumer Price Index (CPI) for the United Arab Emirates, as officially published by the Ministry of Economy / Federal Competitiveness and Statistics Centre. Such adjustment shall apply prospectively to invoices issued after the adjustment date.
- All prices contained in this proposal shall be valid for a period of 30 days from the date of this proposal.
- All prices shown in this proposal shall be paid in AED.
- The price and any other amounts payable for the services and deliverables set out in this proposal are exclusive of any tax, including value-added tax (VAT), which shall be payable by the Client at the prevailing rate.

9.2 Payment Terms

All prices contained in this Proposal shall be subject to the payment terms which are set out in the table below:

Service	Payment Terms
PaaS/IaaS, O&M	Annually in Advance

Table 4: Payment Terms

10 Terms and Conditions

10.1 Proposal

This Proposal is provided for informational purposes only and does not constitute, in whole or in part, a contract for the products and/or services described herein or an offer to enter into a contract.

Core42 makes no representations or warranties, express or implied, with respect to any of the information contained in this Proposal.

To avoid doubt, this Proposal and any correspondence that follows will not be deemed to be included nor form part of the Agreement and/or Work Order to be agreed between the parties.

10.2 Terms of Agreement

In the event that Core42 is selected to carry out the services described within this Proposal, the terms and conditions of the contract will be based on the current Master Service Agreement (MSA).

Further, on notice of award, Core42 will develop and submit a **“Work Order”**. The Work Order will describe the commercial and granular scope elements of this Proposal. Once agreed, the Work Order and the Agreement shall constitute the entire agreement between the parties for the services described in this Proposal and shall supersede this Proposal and any other written or oral correspondence or communication relation to the subject matter of this Proposal. Appendices

11 Appendix:

The following appendices include supplementary information and detailed documentation to provide a comprehensive overview of our proposal.

Appendix	Title
Appendix A	Definitions and Glossary
Appendix B	Company Profile
Appendix C	Oracle Private Cloud Service Catalogue
Appendix D	Security Guidelines
Appendix E	Networking Requirement Considerations
Appendix F	Networking Solutions
Appendix G	Manageability

Table 5: Appendices

11.1 Appendix A: Definitions and Glossary

Term	Definition
TCO	Total Cost of Ownership
CSC	Cyber Security Council
ADGOV	Abu Dhabi Government
UAE IA	UAE Information Assurance
ADISS	Abu Dhabi Information Security Standards
DESC	Dubai Electronic Security Centre
SSH	Secure Shell
AES	Advanced Encryption Standard
TLS	Transport Layer Security
HSM	Hardware Security Module
PII	Personally Identifiable Information
Oracle Private Cloud/DRCC	Dedicated Region Cloud at Customer
BYOL	Bring Your Own License
COA	Chart of Accounts
TOM	Target Operating Model
ETL	Extract, Transform, and Load
OCI	Oracle Cloud Infrastructure
VCN	Virtual Cloud Network

Term	Definition
DBA	Database Administrator
LLM	Large Language Model
VPN	Virtual Private Network
U2F	Universal 2nd Factor
SLA	Service Level Agreement
SOW	Statement of Work
ERP	Enterprise Resource Planning
EPM	Enterprise Performance Management
GRC	Governance, Risk, and Compliance
SCM	Supply Chain Management
HCM	Human Capital Management
ODA	Oracle Digital Assistant
SIT	System Integration Testing2
GRC	Governance, Risk, and Compliance

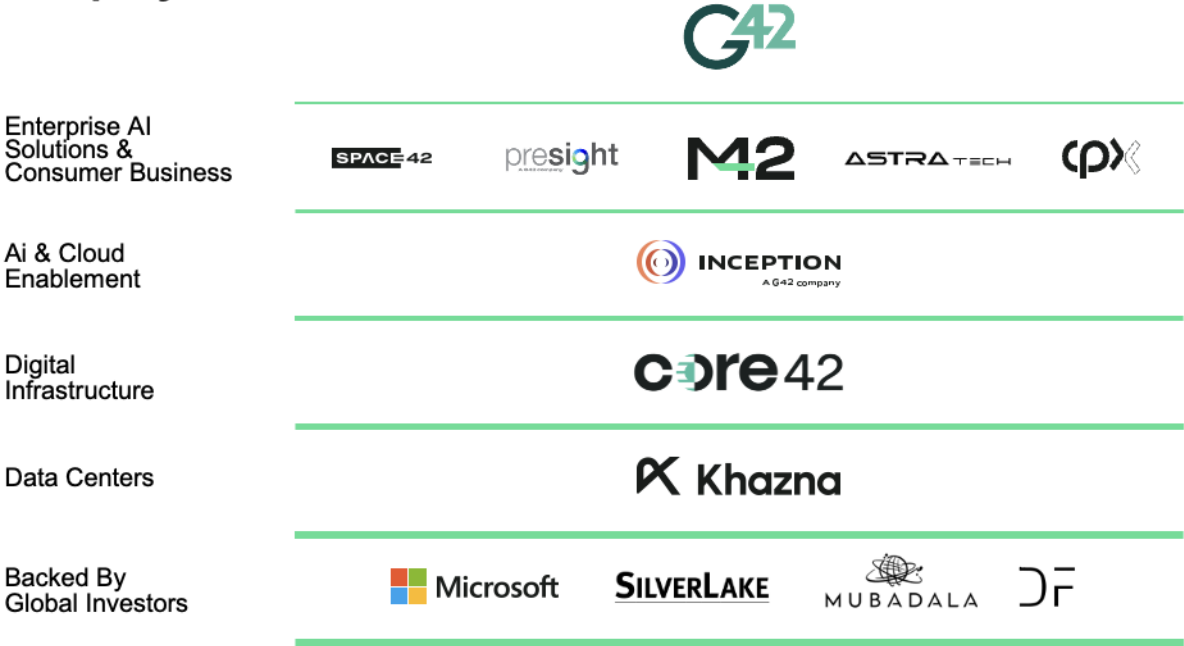
Table 6: Definitions and Glossary

11.2 Appendix B: Company Profile

11.2.1 Our Company Overview

Born in Abu Dhabi and expanding globally

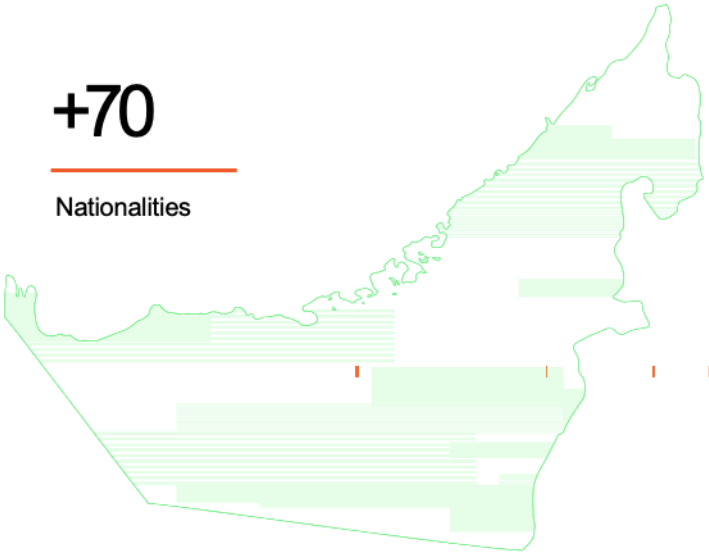
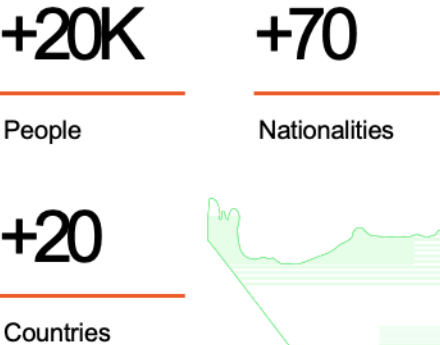
Company structure



G42 Mission

The Intelligence Grid delivers augmented intelligence, on-demand AI services, and limitless capabilities for businesses, governments, and individuals.

Since its start in 2018, **CORE42** has experienced remarkable growth, becoming a global community of:



11.2.2 Our Company Organisation Chart

Leading with purpose. The Core42 Executive Team



Talal M. Al Kaissi
CEO (Interim), Core42
Group Chief Global Affairs
Officer, G42



Sherif Tawfik
Chief Business Officer



Neha Gupta
Chief Financial Officer



Raghu Chakravarthi
EVP, Engineering
and GM Core42 US



Roopal Jobanputra
General Counsel



Rajeev R Nair
SVP - Special Projects



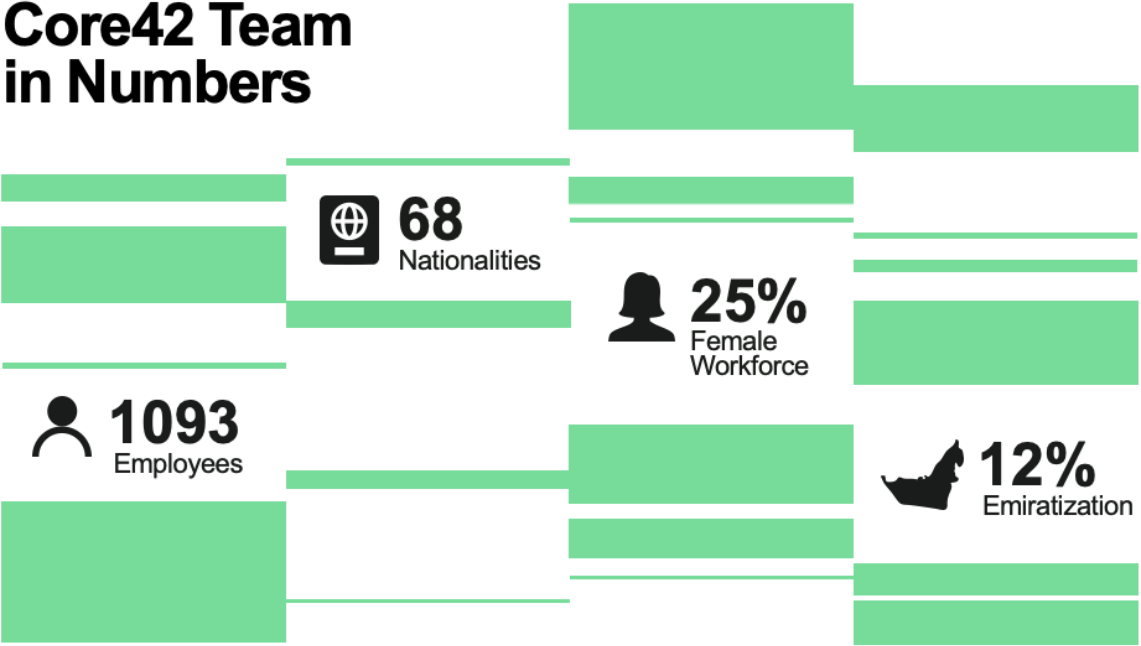
Ergul Hemmingsen
Chief Human Capital Officer

11.2.3 Our Partners and Numbers

Strong partnership ecosystem
with the world’s leading
organizations



Core42 Team
in Numbers



11.3 Appendix C: Oracle Private Cloud Service Catalogue

The **Oracle Private Cloud Service** catalogue offers various services to meet DCT's needs, from powerful computing options and scalable storage to secure networking like FastConnect and load balancing. With monitoring, management, and hybrid cloud support tools, it's designed to provide a secure and flexible cloud experience.

11.3.1 Infrastructure as a Service (IaaS)

Service	Description
Compute	
Bare Metal Compute	High-performance, single-tenant compute instances.
Container Engine for Kubernetes	Managed container orchestration for Kubernetes.
Container Registry	A private container registry is used for storing and managing images.
Virtual Machines	Scalable, on-demand virtual machines for various workloads.
Oracle Cloud VMware Solution	Integrated VMware solution for hybrid cloud environments.
Archive Storage	Low-cost, long-term data archiving and storage.
Block Volumes	High-performance, persistent block storage for compute instances.
File Storage	Fully managed, scalable file storage for various workloads.
Local NVMe SSD	High-performance, local NVMe SSD storage for compute instances.
Object Storage	Scalable, durable object storage for data and applications.
Storage Gateway	Hybrid cloud storage solution for on-premises data.
Networking, Connectivity, and Edge Services	
DNS	Scalable, managed DNS service for domain name resolution.
Email Delivery	Reliable, scalable email delivery service for applications.
FastConnect	Dedicated, high-bandwidth connectivity to Oracle Cloud.
Health Checks	Automated health checks for monitoring application performance.
Load Balancing	Scalable, managed load balancing for distributing traffic.
Service Gateway	Secure, managed access to Oracle Cloud services.

Service	Description
Traffic Management	Advanced traffic management for optimising application performance.
Virtual Cloud Network	Software-defined networking for secure, scalable connectivity.
Management and Governance	
Application Performance Monitoring	Real-time monitoring and analytics for application performance.
Logging	Centralised logging and analytics for application and system logs.
Oracle Cloud Observability and Management Platform	Unified platform for monitoring, logging, and analytics.
Monitoring	Real-time monitoring and alerting for system and application performance.
Notifications	Scalable, managed notification service for applications.
Resource Manager	Centralised management and automation for cloud resources.
Tagging	Flexible, scalable tagging for organising and managing cloud resources.
Architecture Centre	Centralised hub for cloud architecture best practices and resources.

Table 7: IaaS Service Description

11.3.2 Platform as a Service (PaaS)

Platform as a Service (PaaS) helps enterprise IT and independent software vendor (ISV) developers rapidly build and deploy rich applications - or extend Oracle Cloud SaaS apps - using an enterprise-grade cloud platform based on the industry's #1 database and application server.

Service	Description
Analytics and Big Data	
Oracle Analytics Cloud	Cloud-based analytics platform for data analysis and visualisation.
Oracle Big Data Service	Fully managed big data service for data processing and analytics.
Oracle Big Data SQL Cloud Service	Cloud-based SQL service for big data analytics.

Service	Description
Oracle Cloud Infrastructure Data Catalogue	Centralised data catalogue for data discovery and governance.
Oracle Cloud Infrastructure Data Flow	Fully managed data processing service for big data workloads.
Oracle Cloud Infrastructure Streaming	Real-time data streaming service for event-driven applications.
Oracle Data Science	Cloud-based data science platform for machine learning and analytics.
Oracle Essbase	Cloud-based business analytics platform for financial planning and analysis.
Application Development	
API Gateway	Fully managed API gateway for secure API management.
Blockchain Platform	Cloud-based blockchain platform for secure and transparent transactions.
Container Engine for Kubernetes	Managed container orchestration for Kubernetes.
Container Pipelines / DevOps	Automated container pipelines for DevOps and continuous integration.
Container Registry	Private container registry for storing and managing images.
Data Science	Cloud-based data science platform for machine learning and analytics.
Developer	Cloud-based developer platform for building and deploying applications.
Digital Assistant	Cloud-based digital assistant platform for conversational AI.
Events Service	Cloud-based event-driven platform for real-time data processing.
Functions	Cloud-based serverless platform for event-driven applications.
Oracle Content and Experience	Cloud-based content management platform for digital experiences.
Visual Builder	Cloud-based visual development platform for building and deploying applications.
Data Management	

Service	Description
Autonomous Data Warehouse	Fully managed data warehouse service for data analysis and reporting.
Autonomous Transaction Processing	Fully managed transaction processing service for real-time data processing.
Database Cloud Service: Bare Metal	Cloud-based database service for bare metal infrastructure.
Database Cloud Service: Virtual Machine	Cloud-based database service for virtual machine infrastructure.
Exadata Cloud Service	Cloud-based Exadata service for high-performance database workloads.
Oracle NoSQL Database Cloud Service	Cloud-based NoSQL database service for big data and IoT workloads.
Emerging Technologies	
Data Science	Cloud-based data science platform for machine learning and analytics.
Digital Assistant	Cloud-based digital assistant platform for conversational AI.
Machine Learning	Cloud-based machine learning platform for predictive analytics.
Enterprise Integration	
Tactical Edge Cloud	Cloud-based edge computing platform for real-time data processing.
API Gateway	Fully managed API gateway for secure API management.
Oracle Data Integration	Cloud-based data integration platform for data integration and governance.
GoldenGate	Cloud-based data integration platform for real-time data replication.
Application Integration	Cloud-based application integration platform for integrating applications.
SOA Cloud Service	Cloud-based SOA platform for service-oriented architecture.
Security and Identity	
Database Security	Cloud-based database security platform for data encryption and access control.

Service	Description
Identity and Access Management	Cloud-based identity and access management platform for secure access control.
Oracle Cloud Infrastructure Vault	Cloud-based key management platform for secure data encryption.
Web Application Firewall (WAF)	Cloud-based web application firewall for secure web application protection.

Table 8: PaaS Service Description

11.4 Appendix D: Security Guidelines

Oracle Security, Identity, and Compliance

Oracle Cloud Infrastructure (OCI) is designed to protect customer workloads with a security-first approach across compute, network, and storage - down to the hardware. It's complemented by essential security services to provide the required levels of security for DCT's most business-critical workloads.

- Security Strategy - To create a successful security strategy and architecture for DCT's deployments on OCI, it's helpful to understand Oracle's security principles and the OCI security services landscape.
- The security pillar capabilities reflect fundamental security principles for architecture, deployment, and maintenance. The best practices in the security pillar, help DCT to define a secure cloud architecture, identify and implement the right security controls, and monitor and prevent issues such as configuration drift.

References

- The Best Practices Framework for OCI provides architectural guidance about how to build OCI services in a secure fashion, based on recommendations in the Best practices framework for Oracle Cloud Infrastructure.
- Learn more about Oracle Cloud Security Practices.
- For detailed information about security responsibilities in Oracle Cloud Infrastructure, see the Oracle Cloud Infrastructure Security Guide.

Compliance and Regulations

Cloud computing is fundamentally different from traditional on-premises computing. In the traditional model, organisations are typically in full control of their technology infrastructure located on premises (e.g., physical control of the hardware, and full control over the technology stack in production). In the cloud, organisations leverage resources and practices that are under the control of the cloud service provider, while still retaining some control and responsibility over other components of their IT solution. As a result, managing security and privacy in the cloud is often a shared responsibility between the cloud customer and the cloud service provider. The distribution of responsibilities between the cloud service provider and DCT also varies based on the nature of the cloud service (IaaS, PaaS, SaaS).

Additional Resources

- Oracle Cloud Compliance - Oracle is committed to helping customers operate globally in a fast-changing business environment and address the challenges of an ever more complex regulatory environment. This site is a primary reference for customers on the Shared Management Model with Attestations and Advisories.
- Oracle Security Practices - Oracle's security practices are multidimensional, encompassing how the company develops and manages enterprise systems, and cloud and on-premises products and services.
- Oracle Cloud Security Practices documents.
- Contract Documents for Oracle Cloud Services.
- OCI Shared Security Model
- OCI Cloud Adoption Framework Security Strategy
- OCI Security Guide
- OCI Cloud Adoption Framework Security chapter

11.5 Appendix E: Networking Requirement Considerations

The below questions help to identify networking requirements:

Application Connectivity

- Does DCT's application need to be exposed to the internet?
- Does DCT's solution on DC (on-prem) need to be connected 24x7 to OCI in a Hybrid model?
 - Site-to-Site IPSEC (Y/N)
 - Dedicated Lines (FC) (Y/N)
- Are there any specific network security requirements for DCT's application? (No internet, encryption, etc, etc)
- Will DCT's application require connectivity to other cloud providers?
 - Site-to-Site IPSEC (Y/N)
 - Dedicated Lines (FC) (Y/N)
- Will DCT's application require inter-region connectivity?
- Are you planning to reuse IP addresses from DCT's on-premises environment in OCI?
- If yes, what steps have DCT taken to ensure IP address compatibility and avoid conflicts?
- How will DCT handle network address translation (NAT) for IP reuse in OCI?
- Will DCT bring DCT's own public IPs to OCI?

DR and Business Continuity

- Does DCT need a Business Continuity/DR Plan to address potential disruptions?
 - Network Requirements (min latency, bandwidth, etc)
 - RPO/RTO values
- What are DCT's requirements regarding Data Replication and Geo-Redundancy (different regions, restrictions, etc.)?
- Are DCT planning to distribute incoming traffic across multiple instances or regions to achieve business continuity?
- What strategies do DCT require to guarantee minimal downtime and data loss, and to swiftly recover from any unforeseen incidents?

High Availability and Scalability

- Does DCT's application require load balancing for high availability and scalability? (y/n)
 - Does DCT's application span around the globe or is regionally located?
 - How do DCT intend to ensure seamless user experiences and consistent connections in DCT's application (session persistence, affinity, etc.)?
 - What are the network Security requirements for traffic management (SSL offloading, X509 certificates management, etc.)?
 - Does DCT's application use name resolutions and traffic steering across multiple regions (Public DNS steering)?

Security and Access Control

- Some of the below questions help DCT to adopt the right sizing and deployment model of the network firewall.
 - Does the customer need to protect traffic from VCN to VCN?
 - Does the customer need to protect traffic from subnet to subnet in the same VCN?
 - When deploying an OCI Network Firewall in a dedicated HUB or secure VCN, do DCT want to protect inter-VCN traffic and/or inter-subnet traffic from within the same VCN?
 - Does the customer need to protect incoming or egressing traffic to the internet?
 - Does the customer need to protect internal traffic (including on premises via IPSEC/FC)?
 - Is the network performance critical?
 - Does the customer have any requirement on network isolation (i.e., internet traffic never traverses or is mixed with internal traffic)?
- Have DCT considered the importance of protecting DCT's web applications from potential cyber threats using a Web Application Firewall (WAF)?

Monitoring and Troubleshooting

- How do DCT plan to monitor DCT's application's network performance in OCI?
- How can DCT proactively address and resolve any potential network connectivity challenges DCT's company might face?
- How do DCT plan to troubleshoot DCT's network connectivity?

11.6 Appendix F: Networking Solutions

OCI Network Firewall

Oracle Cloud Infrastructure Network Firewall is a next-generation managed network firewall and intrusion detection and prevention service for DCT's Oracle Cloud Infrastructure VCN, powered by Palo Alto Networks®.

- Overview
- OCI Network Firewall

OCI Load Balancer

The Load Balancer service provides automated traffic distribution from one entry point to multiple servers reachable from DCT's virtual cloud network (VCN). The service offers a load balancer with DCT's choice of a public or private IP address and provisioned bandwidth.

- Load Balancing
- Overview
- Concept Overview

OCI DNS Traffic Management

Traffic Management helps DCT guide traffic to endpoints based on various conditions, including endpoint health and the geographic origins of DNS requests.

- Concept Overview
- DNS

OCI WAF

Protect applications from malicious and unwanted internet traffic with a cloud-based, PCI-compliant, global web application firewall service.

- Cloud Security Web Application Firewall
- Add WAF to a load balancer

OCI IGW

An internet gateway is an optional virtual router that connects the edge of the VCN with the internet. To use the gateway, the hosts on both ends of the connection must have public IP addresses for routing.

- Managing IGW

OCI Site-to-Site VPN

Site-to-site VPN provides a site-to-site IPSec connection between DCT's on-premises network and DCT's virtual cloud network (VCN). The IPSec protocol suite encrypts IP traffic before the packets are transferred from the source to the destination and decrypts the traffic when it arrives. Site-to-Site VPN was previously referred to as VPN Connect and IPSec VPN.

- Overview IPSec
- Setup IPSec

OCI Fast Connect

FastConnect allows customers to connect directly to their Oracle Cloud Infrastructure (OCI) virtual cloud network via dedicated, private, high-bandwidth connections.

- FastConnect
- Concept Overview

OCI VTAP

A Virtual Test Access Point (VTAP) provides a way to mirror traffic from a designated source to a selected target to facilitate troubleshooting, security analysis, and data monitoring.

- VTAP
- Network VTAP Wireshark

OCI NPA

Network Path Analyzer (NPA) provides a unified and intuitive capability DCT can use to identify virtual network configuration issues that impact connectivity. NPA collects and analyses the network configuration to determine how the paths between the source and the destination function or fail.

- Path Analyzer

OCI DRG (Connectivity Options)

A DRG acts as a virtual router, providing a path for traffic between DCT's on-premises networks and VCNs, and can also be used to route traffic between VCNs. Using different types of attachments, custom network topologies can be constructed using components in different regions and tenancies.

- Managing DRGs
- OCI Pilot Light DR
- Peering VCNs in different regions through a DRG

OCI Oracle Cloud Infrastructure Certificates

Easily create, deploy, and manage Secure Sockets Layer/Transport Layer Security (SSL/TLS) certificates available in Oracle Cloud. In a flexible Certificate Authority (CA) hierarchy, Oracle Cloud Infrastructure Certificates help create private CAs to provide granular security controls for each CA.

- SSL TLS Certificates

OCI Monitoring

DCT can monitor the health, capacity, and performance of DCT's Oracle Cloud Infrastructure resources by using metrics, alarms, and notifications. For more information, see Monitoring and Notifications.

- Networking Metrics

11.7 Appendix G: Manageability

OCI offers a full set of services to cover all Observability and Monitoring requirements.

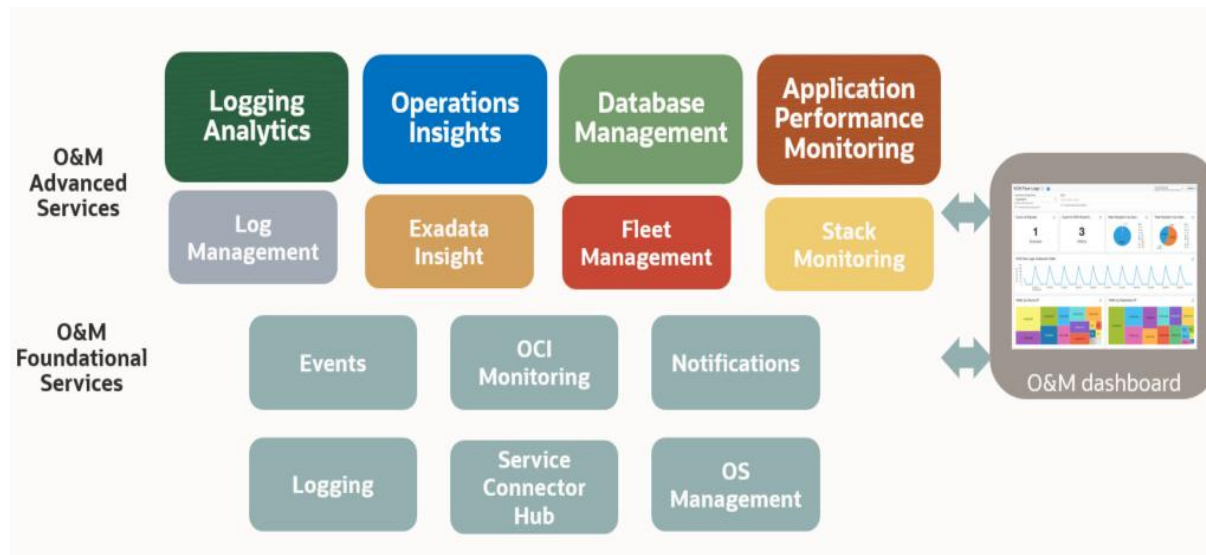


Figure 18: OCI Manageability

OCI Observability

Thanks to AI algorithms the OCI O&M (Observability and Management) solutions offer valuable insights into system status, requirements, and trends. Furthermore, it identifies SQL performance issues. This proactive approach empowers proactive measures to prevent future issues.

OCI O&M Services List

The observability and management services include the following services:

Application Performance Monitoring offers in-depth insight into application performance and facilitates rapid diagnostics to ensure a reliable level of service. This includes monitoring various components and application logic spread across customers, third-party services, and backend computing tiers, whether on-premises or in the cloud.

Database Management provides comprehensive database performance diagnostics and management capabilities to monitor and manage Oracle databases.

Logging lets DCT enable, view, and manage all the logs in DCT's tenancy and provides access to logs from Oracle Cloud Infrastructure resources. These logs include critical diagnostic information that describes how resources are performing and being accessed.

Logging Analytics is a unified, integrated cloud solution that enables users to monitor, aggregate, index, analyse, search, explore, and correlate all log data from their applications and system infrastructure.

OCI Monitoring enables DCT to query metrics and manage alarms. Metrics and alarms help monitor the health, capacity, and performance of DCT's cloud resources.

Ops Insights provides a 360-degree insight into the resource utilisation and capacity of Oracle Autonomous Databases. DCT can easily analyse CPU and storage resources, forecast capacity issues, and proactively identify SQL performance issues across a fleet of Autonomous Databases.

Service Connector Hub is a cloud message bus platform that offers a single pane of glass for describing, running, and monitoring interactions for data moving between Oracle Cloud Infrastructure services.

Stack Monitoring enables proactive monitoring of applications and their underlying stack, including application servers and databases. By discovering all components of an application, including the application topology, Stack Monitoring automatically collects status, load, response, error, and utilisation metrics for all application components. Each component of the application stack is referred to as a resource.

Real-Time Monitoring Annex

Service/Product Name	Description	Collateral
Monitoring	OCI Monitoring collects PaaS and IaaS OCI services metrics. It is enabled by default for all the OCI services.	List of metrics collected by default
OCI Application Performance Monitor	APM is a Distributed Tracing System as a Service. It enables DevOps teams to follow every step of every task. It uses open standards such as OpenTelemetry to monitor various programming languages. Plus, it includes a dedicated Java agent to track older J2EE applications, ensuring complete transaction tracing even in mixed environments.	OCI Application Performance Monitoring
OCI Console	The Service Console offers a list of visual representations and basic information about critical metrics like CPU, memory, and storage.	OCI Console Resource Usage Tracking
OCI Database Management (opt to OEM)	It is an OCI-managed service that simplifies database operations and enhances efficiency. It offers advanced monitoring and diagnostic capabilities, enabling proactive management and optimisation of database performance.	List of metrics collected by OCI Database Management
Stack Monitoring	Stack Monitoring lets DCT proactively monitor an application and its underlying application stack, including application servers and databases.	Stack Monitoring for Oracle Database
Third-Party Tools - Service Connector Hub	OCI provides complete O&M capabilities. However, for customers who prefer to use their own tools, OCI allows seamless integration through the Service Connect Hub.	OCI Connector Hub Third-Party

Service/Product Name	Description	Collateral
		Tools Use Cases

Table 9: Real-Time Monitoring Annex

Performance and Tuning Annex

Service/Product Name	Description	Collateral
OCI Logging	The OCI Logging service is a highly scalable and fully managed single pane of glass for all the logs in DCT's tenancy. Logging provides access to logs from Oracle Cloud Infrastructure resources. These logs include critical diagnostic information that describes how resources are performing and being accessed.	OCI Logging
OCI Monitoring	Use the Oracle Cloud Infrastructure Monitoring service to actively and passively monitor cloud resources using the Metrics and Alarms features. Metric data posted to the Monitoring service is only presented to DCT or consumed by the Oracle Cloud Infrastructure features that DCT enables to use metric data.	OCI Monitoring
OCI Dashboard	The Console Dashboards service allows DCT to create custom dashboards in the Oracle Cloud Infrastructure Console to monitor resources, diagnostics, and key metrics for DCT's tenancy.	OCI Dashboard
OCI Logging Analytics	OCI Logging Analytics empowers users to analyse log data from diverse sources across their infrastructure. It provides insights into system performance, identifies trends, and enables proactive resource optimisation by correlating data from multiple layers of the infrastructure.	OCI Logging Analytics
OCI Application Performance Monitor	APM allows to drill down from user sessions till the single DB query or external call to identify performance bottleneck.	OCI Application Performance Monitoring
OCI Database Management - PerfHub	Is an OCI-managed service that offers performance and tuning capabilities. It provides the same performance and tuning features as the Oracle Enterprise Manager (OEM) Performance and Tuning Pack but in a managed solution.	Database Management Performance Hub

Service/Product Name	Description	Collateral
Ops Insights SQL Warehouse and Capacity Planning	OCI Ops Insights allows for the tracking of metrics charts and data collection. It allows for the correlation of resources across various infrastructure layers. Additionally, it predicts high resource utilisation for computing and database instances.	OCI Operations Insight SQL Warehouse OCI Operations Insight Capacity planning

Table 10: Performance and Tuning Annex

Administration Annex

Service/Product Name	Description	Collateral
OCI Console	The OCI Console is embedded in all cloud services. It allows basic tasks such as listing, starting, stopping, or termination of resources.	OCI Console
OCI Database Management	This OCI-managed service allows DCT to manage DCT's databases. It provides a subset of functionalities offered by the OEM.	Database Management
OCI Organisation Management	The OCI Console has several tenancy management features. DCT can use Organisation Management to centrally manage DCT's multi-tenancy environment.	Organisation Management

Table 11: Administration Annex

Troubleshooting Annex

Service/Product Name	Description	Collateral
Logging Analytics	OCI Logging Analytics can handle log events generated by all software applications and infrastructure on the cloud or on premises. For Oracle software logs, a predefined severity pre-classification exists based on Oracle experience.	OCI Logging Analytics OCI Logging Analytics for Exa
OCI Application Performance Monitor	APM allows user to drill down from user sessions till the application logs to find the root cause.	OCI Application Performance Monitoring
OCI Database Management	OCI-managed service that allows DCT to drill down and correlate metrics and data from different layers. It provides built-in links that allow DCT to connect to other O&M services (ex. Ops Insights).	Database Management

Service/Product Name	Description	Collateral
Ops Insights	OCI Ops Insights allows tracking of metrics charts and data collection. It allows for the correlation of resources from different infrastructure layers.	OCI Operations Insight OCI ExaInsight

Table 12: Troubleshooting Annex

Cost Control and Chargeback Annex

Service/Product Name	Description	Collateral
Cost Analysis	Cost Analysis is an easy-to-use visualisation tool to help DCT track and optimise DCT's Oracle Cloud Infrastructure spending. It allows for the generation of charts and the download of accurate and reliable tabular reports of aggregated cost data. With tags, DCT can associate the forecast and the consumption to a specific department.	OCI Cost Analysis
Usage Rest API	OCI offers various Rest APIs to manage services, including the one for cost management.	OCI Usage Rest API

Table 13: Cost Control and Chargeback Annex

